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AIUB, Data Communication Notes

DATA COMMUNICATION

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INTRODUCTION TO DATA COMMUNICATION

Characteristics of Data Communication

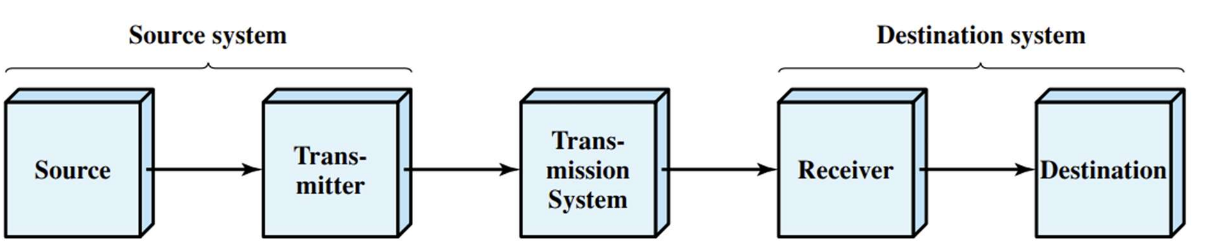
➔ Data Communications:

- Exchange of data between devices via a transmission medium (e.g., wire cable).
- Requires hardware (physical equipment) and software (programs).
- **Effectiveness depends on:**
    1. **Delivery** – Data must reach the correct destination.
    2. **Accuracy** – Data must be transmitted without errors.
    3. **Timeliness** – Data must be delivered on time.
    4. **Jitter** – Variation in packet arrival time should be minimal.

Key Elements of Data Communication

➔ Elements of Data Communication:

1. **Source** – Generates data (e.g., telephone, PC).
2. **Transmitter** – Converts data into transmittable signals (e.g., modem).
3. **Transmission System** – Medium for data transfer (e.g., transmission line, network).
4. **Receiver** – Converts received signals back into data (e.g., modem).
5. **Destination** – Receives and processes incoming data.



(a) General block diagram

A Data Communications Model

➔ Message Transmission Process in a PC:

1. **Message Entry** – User types of message in an email application via keyboard (input device).
2. **Buffering** – Message is stored in main memory as a bit sequence **g**.
3. **Connection to Medium** – PC connects to a network or telephone line via an I/O (Input/ Output) device (transmitter).
4. **Voltage Shifts** – Input data are sent to the transmitter as voltage shifts **g(t)**.
5. **Signal Conversion** – Transmitter converts **g(t)** into a transmission signal **s(t)** suitable for the medium.
6. **Signal Impairment** – The transmitted signal s(t) may be distorted in the medium, resulting in r(t) at the receiver.
7. **Signal Estimation** – The receiver estimates the original s(t) from r(t), producing a bit sequence g'(t).
8. **Bit Transfer** – The recovered bits g'(t) are sent to the destination PC as a block of bits.
9. **Error Detection & Correction** – The destination system checks for errors and, if needed, cooperates with the source for error-free data recovery.
10. **Message Output** – The final message m' is displayed to the user via a screen or printer, usually matching the original m.

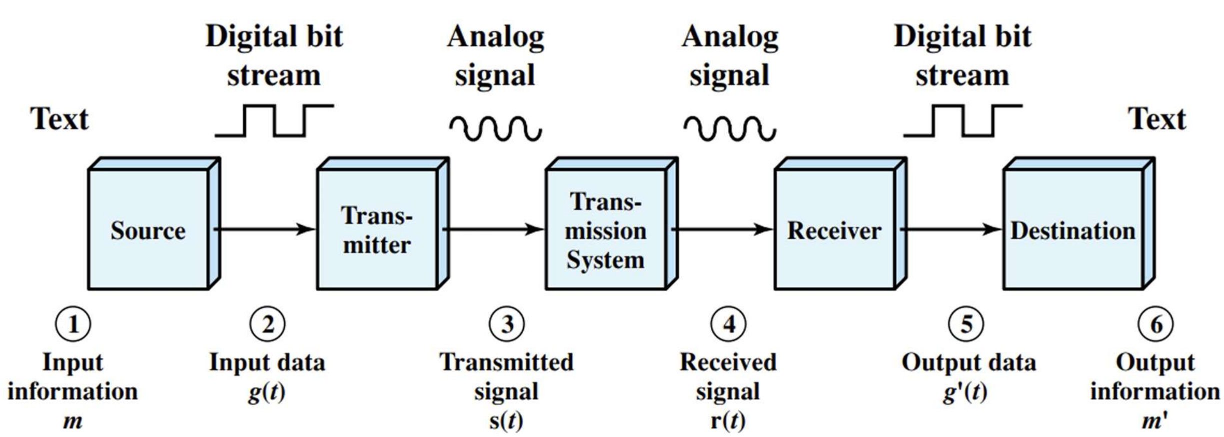


Figure: A Data Communications Model

Data Representation

➔ Binary Representation & Code Sets:

- A **bit** (binary digit) has two states: **0** and **1**.
- Computers require more symbols than a single bit can represent.
- Symbols include:
  ➢ **52 letters** (uppercase & lowercase)
  ➢ **10 numerals** (0–9)
  ➢ **Punctuation & special characters**
  ➢ **Control characters** (e.g., Carriage Return, Line Feed)

➔ Bit Groups & Symbol Representation:

- **5-bit code** → 2^5= 32 symbols.
- **8-bit code** → 2^8= 256 symbols.
- A **code set** assigns bit patterns to symbols.

⇨ Common Code Sets:

1. **ASCII (7-bit)** – American Standard Code for Information Interchange.
2. **EBCDIC (8-bit)** – Extended Binary-Coded-Decimal Interchange Code (IBM).

ASCII Code

➔ ASCII Code Set:

- **Most common & widely used** code set.
- **7-bit code → 128 possible symbols** with defined meanings.

➔ ASCII Symbol Categories:

1. **96 Graphic Symbols** (Columns 2–7):
   - **94 Printable Characters** (letters, digits, punctuation, etc.).
   - **SPACE & DELETE characters**.
2. **32 Control Symbols** (Columns 0 & 1):
   - Used for **text formatting & device control** (e.g., Carriage Return, Line Feed).

7-Bit ASCII (American Standard Code for Information Interchange)								
	000...	001...	010...	011...	100...	101...	110...	111...
..0000	*NUL*	*DLE*	*SP*	0	@	P	‘	p
..0001	*SOH*	*DC1*	!	1	A	Q	a	q
..0010	*STX*	*DC2*	"	2	B	R	b	r
..0011	*ETX*	*DC3*	#	3	C	S	c	s
..0100	*EOT*	*DC4*	$	4	D	T	d	t
..0101	*ENQ*	*NAK*	%	5	E	U	e	u
..0110	*ACK*	*SYN*	&	6	F	V	f	v
..0111	*BEL*	*ETB*	’	7	G	W	g	w
..1000	*BS*	*CAN*	(	8	H	X	h	x
..1001	*HT*	*EM*	)	9	I	Y	i	y
..1010	*LF*	*SUB*	*	:	J	Z	j	z
..1011	*VT*	*ESC*	+	;	K	[	k	{
..1100	*FF*	*FS*	,	<	L	\	l	|
..1101	*CR*	*GS*	-	=	M	]	m	}
..1110	*SO*	*RS*	.	>	N	^	n	~
..1111	*SI*	*US*	/	?	O	_	o	*DEL*

Example 1:

➔ To determine the **binary representation** of a character from its **hexadecimal coordinate**:

1. The first hex digit represents the **row** (4 in this case).
2. The second hex digit represents the **column** (B in this case).
3. Convert both hex digits to **binary** separately:
   - **4 (Hex) → 100 (Binary)**
   - **B (Hex) → 1011 (Binary)**
4. Combine both to get the **7-bit ASCII code**:
   - **K (4B) → 100 1011 (Binary)**

This method works for any ASCII character with a given **hex coordinate**.

Example 2:

- **ASCII Representation of "3P.bat":**
- Each character is represented using its **7-bit ASCII code**, with the **8th bit set to 0**:

Character	Hex Code	Binary (7-bit)	8-bit (MSB = 0)
3	**33**	011 0011	**0011 0011**
P	**50**	101 0000	**0101 0000**
.	**2E**	010 1110	**0010 1110**
b	**62**	110 0010	**0110 0010**
a	**61**	110 0001	**0110 0001**
t	**74**	111 0100	**0111 0100**

Final ASCII Binary Representation:

`00110011 01010000 00101110 01100010 01100001 01110100`

This is the **8-bit ASCII** representation of `"3P.bat"`.

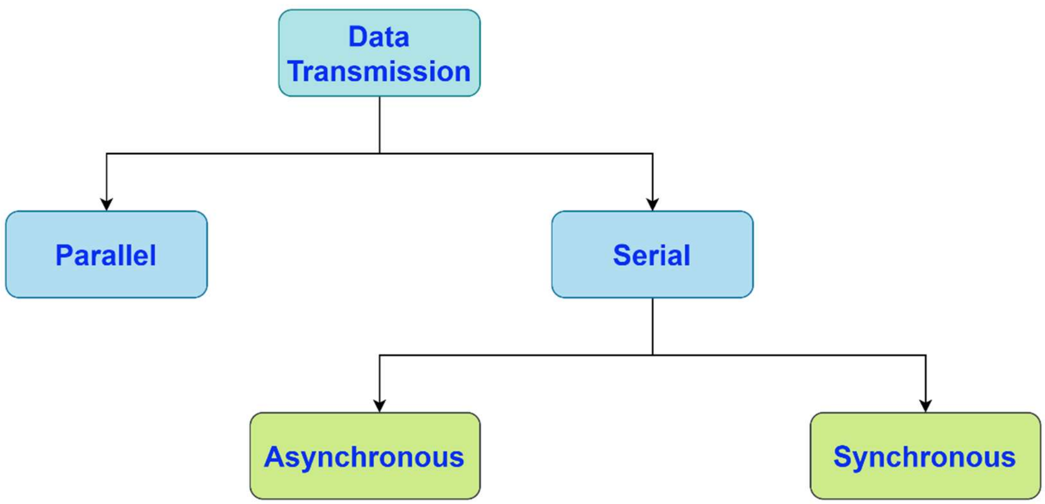
Data Transmission

Bytes & Data Transmission:

- **Byte**: A **group of bits** treated as a single unit during processing.
  - Typically, **8 bits long**, but can vary.
  - Example: **1001011 (ASCII for 'K')** is a byte with a defined meaning.
  - Some bytes **may not belong** to standard code sets.
- **Need for Data Exchange**:
  - Computers and terminals **exchange data, commands, and control information** in binary form.
- **Data Transmission**:
  - Movement of **bits** over a **physical medium** connecting digital devices.

Types of Data Transmission:

1. **Parallel Transmission**
   - Multiple bits are sent **simultaneously** over multiple channels.
   - **Faster** but requires more wires.
   - Used for **short-distance communication** (e.g., inside a computer, printer connections).
2. **Serial Transmission**
   - Bits are sent **one after another** over a single channel.
   - **Slower** but requires fewer wires, making it **cost-effective**.
   - Used for **long-distance communication** (e.g., USB, Ethernet, Internet).



Serial Synchronous Transmission

- **Continuous transmission** of bits without start/stop bits or gaps.
- The **receiver** groups the bits based on timing.
- **Faster** and more **efficient** than asynchronous transmission.
- Used in **high-speed communication** (e.g., Ethernet)

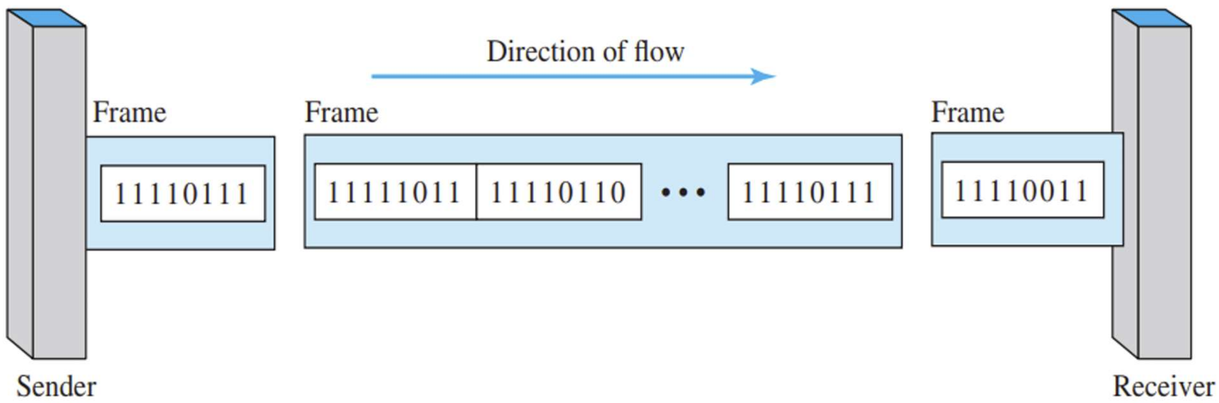


Figure: Synchronous transmission

Example 1:

➔ Write the Synchronous bit transmission sequence of the message "HI".

Character	Binary (7-bit)	8-bit (MSB = 0)	After transmission
H	1001000	01001000	00010010
I	1001001	01001001	10010010

Example 2:

➔ Write the Synchronous bit transmission sequence of the message "3P.bat".

Character	Binary (7-bit)	8-bit (MSB = 0)	After transmission
3	0110011	00110011	11001100
P	1010000	01010000	00001010
.	0101110	00101110	01110100
b	1100010	01100010	01000110
a	1100001	01100001	10000110
t	1110100	01110100	00101110

Serial Asynchronous Transmission

- **Add Start Bit (0)** at the beginning of each byte and **stop bits (1)** at the end.
- **Variable gaps** between bytes (also called framing).
- **Used for low-speed data transmission**.
- Each byte is sent **individually** with timing information provided by the start and stop bits.

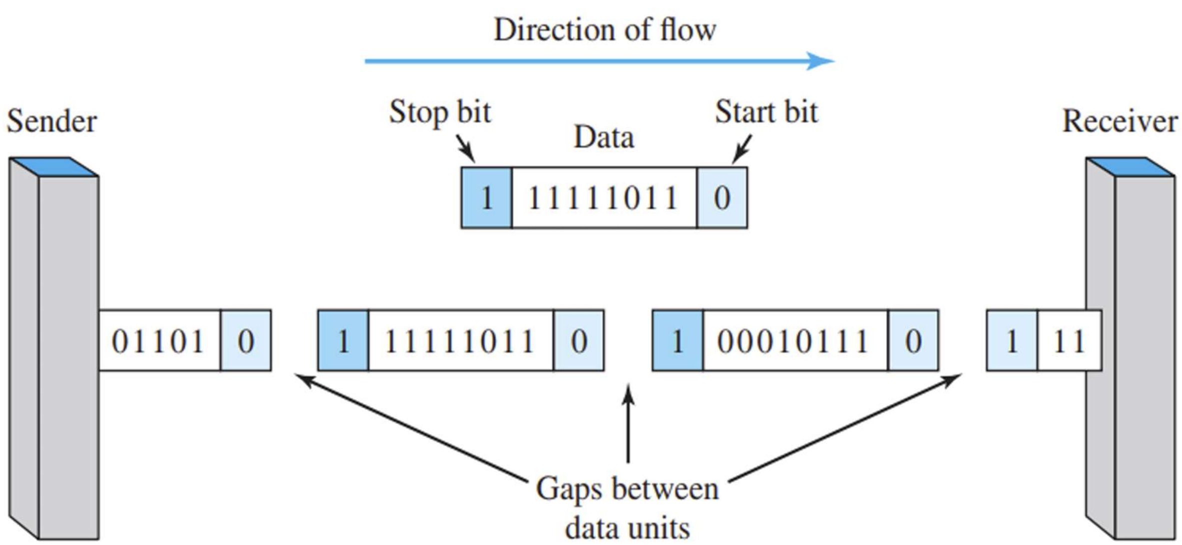


Figure: Asynchronous transmission

Example 1:

➔ Write the Asynchronous bit transmission sequence of the message "HI".

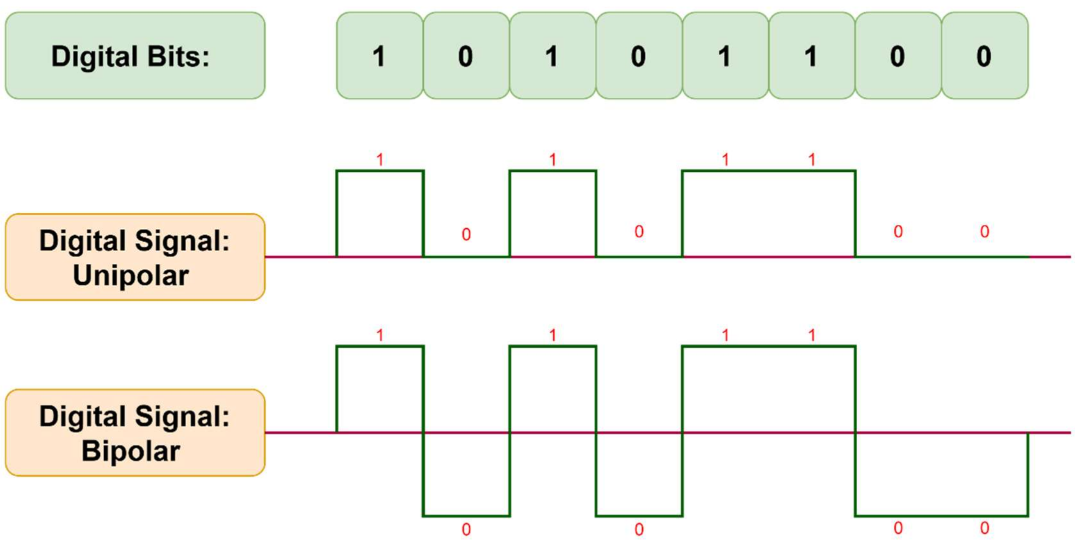
Character	Binary (7-bit)	8-bit (MSB = 0)	Add Start & Stop Bit	After transmission
H	1001000	01001000	1010010000	0000100101
I	1001001	01001001	1010010010	0100100101

Bit Representation in Transmission

- **Electrical signals** are used to transmit bits over wires.
- **Two binary states ("1" and "0")** are represented by different voltage levels.

➔ Types of Transmission:

1. **Unipolar Transmission:**
    - One state (e.g., **"0"**) is **0 volts**.
    - The other state (e.g., **"1"**) is a **positive voltage (+V volts)**.
2. **Bipolar Transmission:**
    - **"1"** is represented by **+V volts**.
    - **"0"** is represented by **-V volts**.
    - More efficient in **reducing signal degradation** over long distances.



Bit Rate

- **Bit Rate (R)**: Number of **bits transmitted per second**.
- **Formula**: $\boldsymbol{R = \frac{1}{tp}}$ where t_p is the **bit duration**.
- **Bit Duration ≠ Pulse Duration**:
  - A **pulse** can span multiple bit durations.
  - In some signal formats, **pulse duration** may be **half** of the bit duration.

Types of Networks

1. **Local Area Network (LAN)**
   - Covers a **small area** (e.g., home, office, school).
   - High **data transfer speed** and low latency.
   - Example: **Wi-Fi, Ethernet** in a building.
2. **Metropolitan Area Network (MAN)**
   - Covers a **city or large campus**.
   - Connects multiple **LANs** within a **metropolitan area**.
   - Example: **Cable TV networks, city-wide Wi-Fi**.
3. **Wide Area Network (WAN)**
   - Covers **large geographical areas** (e.g., countries, continents).
   - Slower than LAN/MAN but supports **long-distance communication**.
   - Example: **The Internet, satellite networks**.

NETWORK MODELS

Layered Tasks

➔ Protocol Layering:

- **Concept of Layers**: Used in daily life to **break down complex tasks** into **smaller, manageable steps**.
- **Why Use Layers?**
  - **Simplifies network design** by handling specific tasks at each layer.
  - **Enhances troubleshooting** since issues can be isolated to a single layer.
  - **Promotes compatibility** between different systems.
- **Example**: The **OSI model** and **TCP/IP model** use layering to structure network communication.

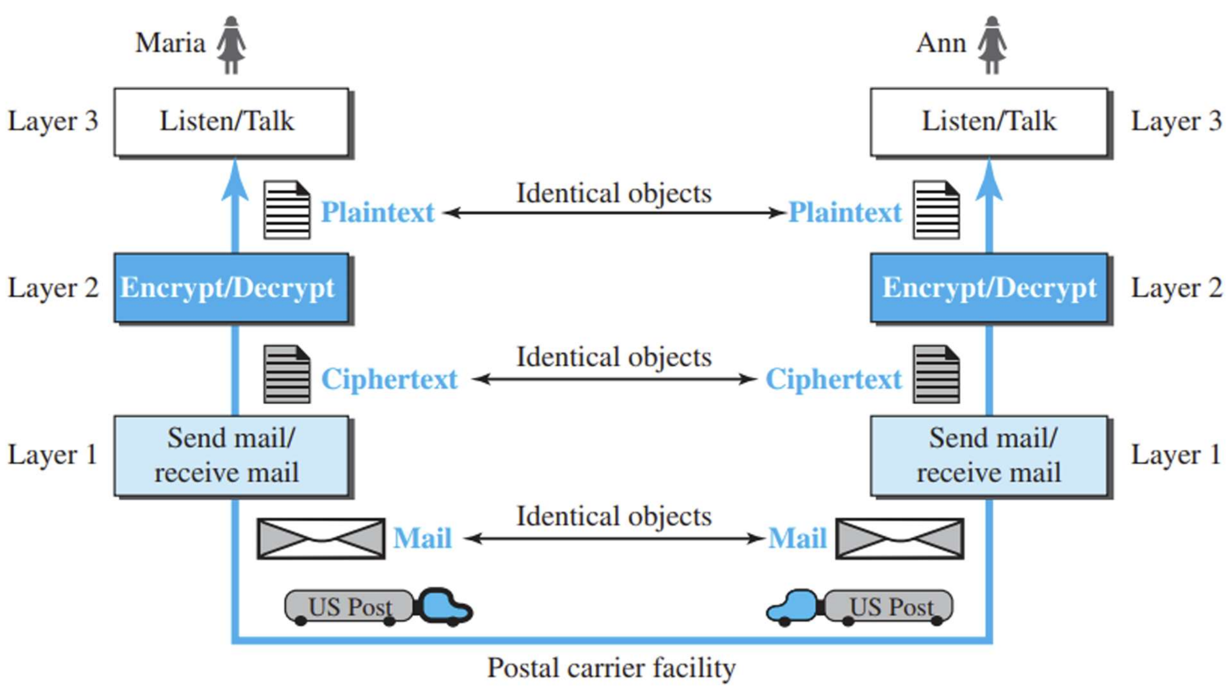


Fig: Tasks involved in sending a message in multiple layer

Importance of Layering

➔ Advantages of Protocol Layering:

1. **Troubleshooting**
   - Each layer has specific **protocols, actions, and data**.
   - Issues can be **pinpointed** to the layer responsible for the problem.
2. **Standards & Interoperability**
   - Provides a **structured guideline** for different vendors.
   - Ensures that **devices from different manufacturers** can communicate seamlessly.
3. **Flexibility & Change Management**
   - **Changes in one layer** have **minimal impact** on others.
   - Prevents disruptions across the entire system.

Introduction to OSI Model

➔ OSI Model – Seven Layers

The **Open Systems Interconnection (OSI) Model** is an **abstract framework** for network communication, developed by **ISO in 1978**. It consists of **seven layers**, each handling specific network functions.

OSI Seven Layers (Top to Bottom):

1. **Application Layer** – User interaction (e.g., web browsers, email).
2. **Presentation Layer** – Data formatting, encryption, compression.
3. **Session Layer** – Manages connections (start, maintain, terminate).
4. **Transport Layer** – Ensures reliable data delivery (TCP, UDP).
5. **Network Layer** – Handles routing and addressing (IP, routers).
6. **Data Link Layer** – Manages data frames and MAC addressing.
7. **Physical Layer** – Deals with actual transmission (cables, signals).

OSI Model	
Data unit	Layer
Data	7. Application
	6. Presentation
	5. Session
Segments	4. Transport
Packet	3. Network
Frame	2. Data Link
Bit	1. Physical

Application Layer

➔ Application Layer (Layer 7) – OSI Model

- **Closest to the end user**, directly interacting with software applications.
- Provides **network services** like web browsing, file transfer, and email communication.

Examples of Application Layer Protocols:

1. **HTTP (Hypertext Transfer Protocol)** – Used for **web browsing** (e.g., accessing websites).
2. **FTP (File Transfer Protocol)** – Facilitates **file transfers** between computers.
3. **SMTP (Simple Mail Transfer Protocol)** – Handles **email transmission**.

This layer ensures seamless communication between **user applications and network services**.

Presentation Layer

➔ Presentation Layer (Layer 6) – OSI Model

- **Role**: Ensures **interoperability** between different encoding methods used by computers.
- **Responsibilities**:
  - Translates data between the **network format** and the **computer format**.
  - Handles **data representation differences**, such as **encryption** and **compression**.

Session Layer

➔ Session Layer (Layer 5) – OSI Model

- **Controls dialogue** between two end systems (manages sessions).
- Defines how to **start, control, and end** conversations.
- Handles **log-on** and **password validation**.
- Responsible for **terminating** the connection.

Transport Layer

➔ Transport Layer (Layer 4) – OSI Model

- **Purpose**: Provides reliable data exchange between processes on different computers.
- **Responsibilities**:
  - Ensures **error-free** data delivery.
  - Ensures data is delivered in **sequence**.
  - Prevents **data loss** or **duplication**.
  - Provides **connectionless** or **connection-oriented services**.
  - Manages **connection establishment** and termination.
  - **Multiplexes** multiple connections over a single channel.

Network Layer

➔ Network Layer (Layer 3) – OSI Model

- **Routing**: Determines the **best route** for data to travel from source to destination.
- **Logical Addressing**:
  - Defines an **addressing scheme** (e.g., **IP addresses**) to uniquely identify devices on the internetwork.
  - Places the **sender** and **receiver's IP addresses** in the header to distinguish devices uniquely.

Data Link Layer

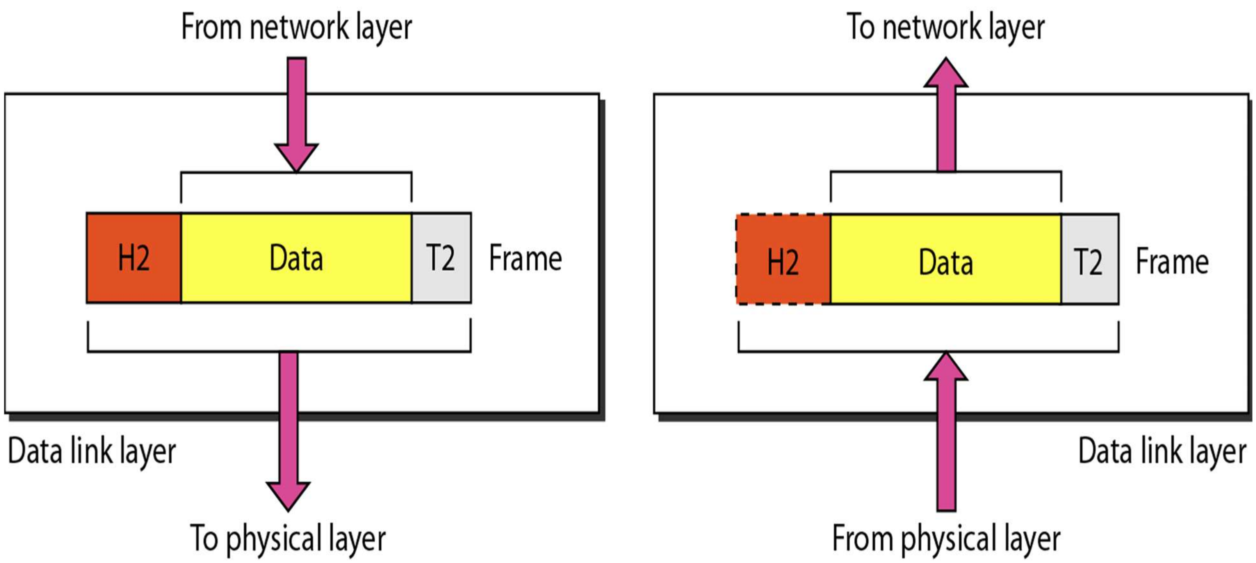
➔ Data Link Layer (Layer 2) – OSI Model

- **Function**: Ensures **node-to-node** error-free data transfer over the **physical layer**.
- **Sub-layers**:
  1. **MAC (Media Access Control)**:
     - Manages how data is **transferred over the media** (e.g., CSMA/CD, CSMA/CA).
  2. **LLC (Logical Link Control)**:
     - Controls **synchronization, flow control**, and **error checking** functions.

⇨ Services of Data-Link Layer:

- **Framing**: Divides the bit stream into **manageable data units** called **frames**.
- **Physical Addressing**: Adds a **header** to define the physical address (MAC address) of the sender/receiver.
- **Flow Control**: Ensures the receiver isn't overwhelmed by controlling the rate of data transfer.
- **Error Control**: Provides **reliability** by detecting and retransmitting lost or damaged frames.
- **Access Control**: Determines **which device controls the link** when multiple devices share the same connection.

➔ Data Link Layer - Frame:



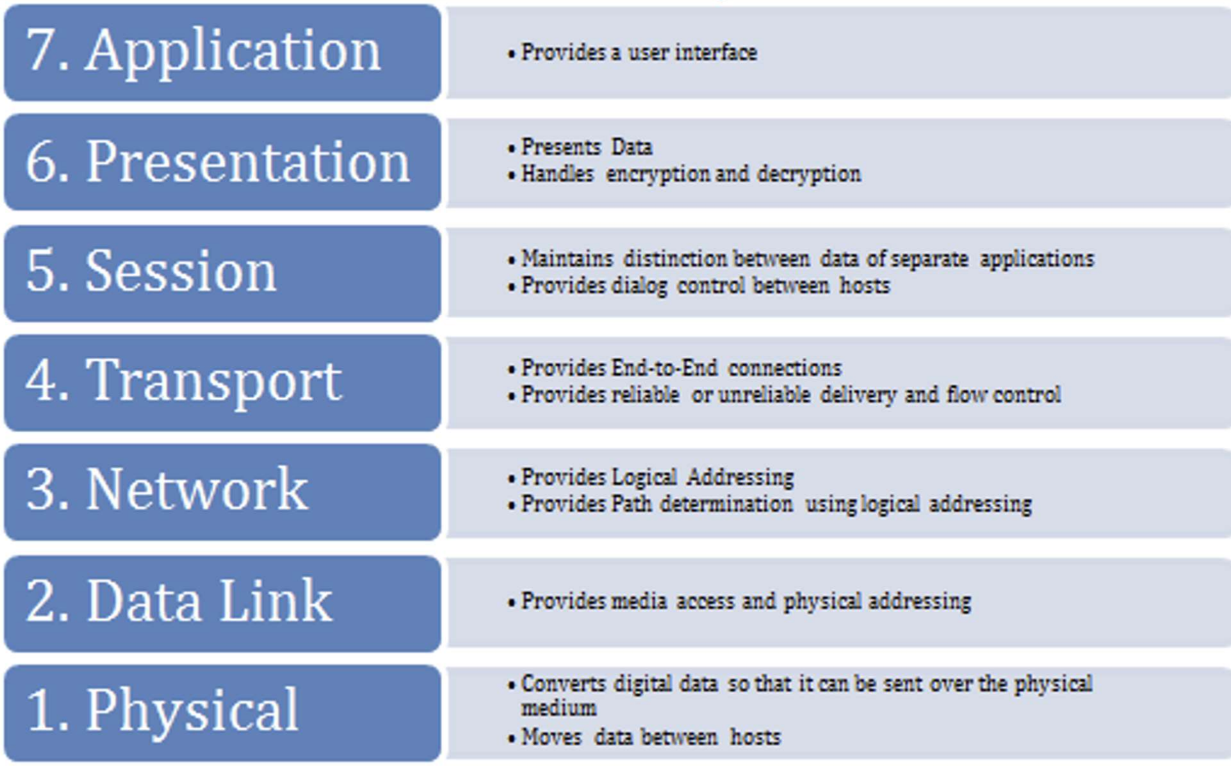
The data link layer is responsible for moving frames from one hop (node) to the next

Physical Layer

➔ Physical Layer (Layer 1) – OSI Model

- **Bit Synchronization**: Provides synchronization of bits using a **clock** to control both sender and receiver.
- **Bit Rate Control**: Defines the **transmission rate** (bits per second).
- **Physical Topologies**: Specifies the physical arrangement of devices (e.g., **bus**, **star**, **mesh** topologies).
- **Transmission Modes**: Defines how data flows between devices:
    - **Simplex**: One-way transmission.
    - **Half-Duplex**: Two-way transmission, but not simultaneously.
    - **Full-Duplex**: Two-way transmission simultaneously.

Summary



Introduction to TCP/IP Model

➔ TCP/IP (Internet Protocol Suite):

- **Definition**: A set of communication protocols used for the Internet and similar networks.
- **Key Protocols**: Named after **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**.
- **Function**: Enables data transmission and communication between networked devices.
- **Standardization**: One of the earliest networking protocol standards.
- **Usage**: Forms the foundation of modern internet communication.

TCP/IP Model in relation to OSI Model

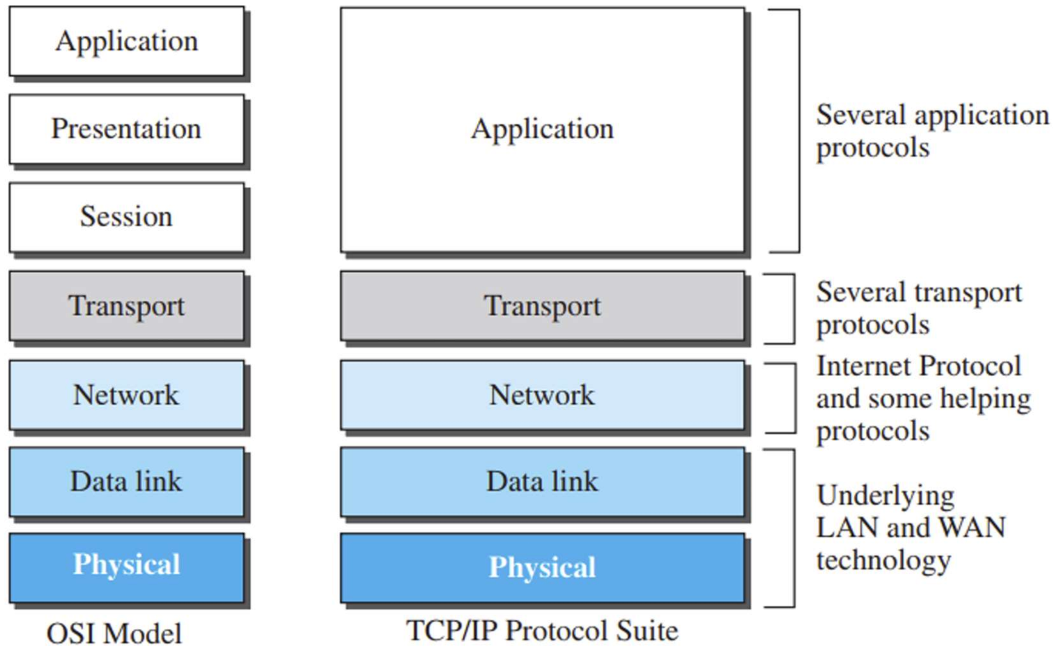


Fig: TCP/IP and OSI model

Data Encapsulation

Encapsulation in the TCP/IP Model

1. **Application Layer**: Sends the email data to the Transport layer.
2. **Transport Layer**: Adds its own header (e.g., port information) and passes data to the Internet layer.
3. **Internet Layer**: Adds a header (e.g., source & destination IP addresses) and forwards data to the Network Access layer.
4. **Network Access Layer**: Adds both a header and a trailer before sending data through the physical network.

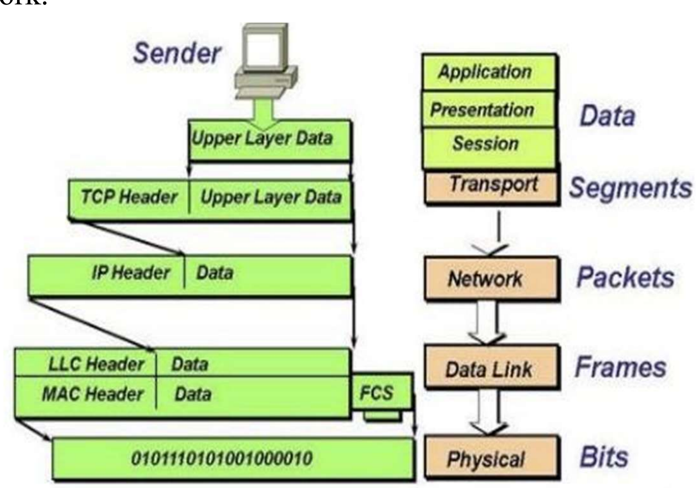


Figure: Data Encapsulation

Data Decapsulation

➔ Decapsulation in the TCP/IP Model

- **Definition**: The reverse of encapsulation, where data is unpacked as it moves up the layers.
- **Process Flow**: Starts from the **Physical Layer** and moves up to the **Application Layer**.
- **Physical Layer**: Receives raw data and passes it to the **Network Access Layer**.
- **Network Access Layer**: Removes its header and trailer, then sends data to the **Internet Layer**.
- **Internet & Transport Layers**: Sequentially strip their headers and forward data upward.
- **Application Layer**: Receives the original data for processing by the intended program.

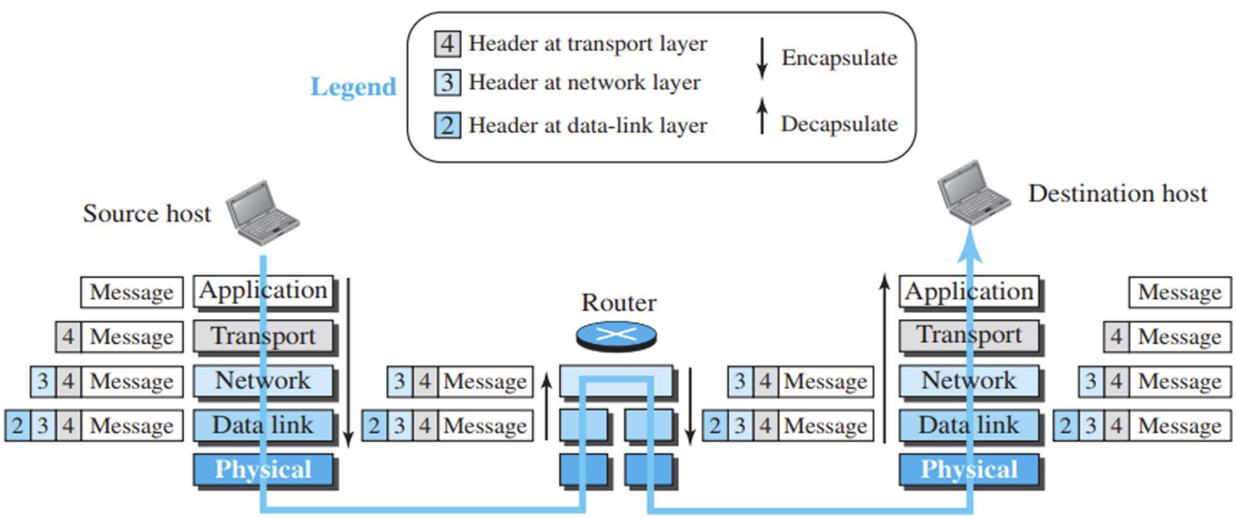


Figure: Encapsulation and decapsulation process

ANALOG & DIGITAL SIGNALS

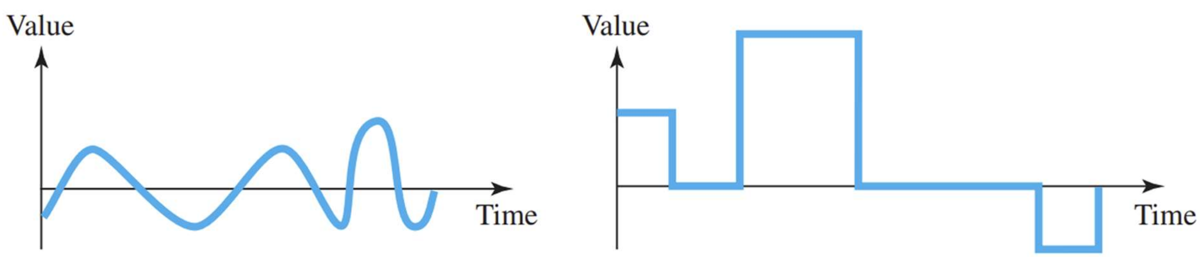
Analog & Digital Signals

➪ **Data Types:**

- **Analog Data**: Continuous information
  - Example: Voice, video, sensor data (temperature, pressure)
- **Digital Data**: Discrete states
  - Example: Computer memory (0s and 1s), text, integers

➪ **Signal Types:**

- **Analog Signals**: Infinite number of values in a range
- **Digital Signals**: Limited number of values
-



a. Analog signal b. Digital signal

Periodic and Non-periodic Signals

➔ **Signal Forms:**

- **Periodic**
- **Nonperiodic (Aperiodic)**

❖ **Periodic Signal:**

- Completes a pattern within a measurable time frame (period)
- Repeats the pattern over identical periods
- One full pattern = **Cycle**

❖ <u>**Nonperiodic Signal:**</u>

- Changes without a repeating pattern or cycle

➔ **Analog & Digital Signals:**

- Both can be periodic or nonperiodic
- **Data Communications**:
  - **Periodic <u>Analog</u> Signals** are common
  - **Nonperiodic <u>Digital</u> Signals** are common

Periodic Analog Signals

➔ Periodic analog signals can be classified as

- **Simple**: Cannot be <u>decomposed</u> (Sine wave)
- **Composite**: Composed of multiple sine waves

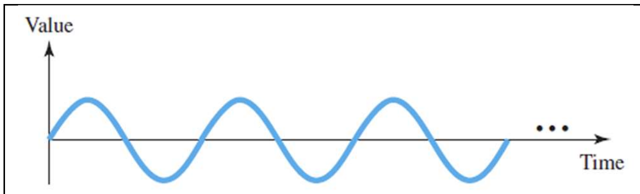


Fig: Simple signal

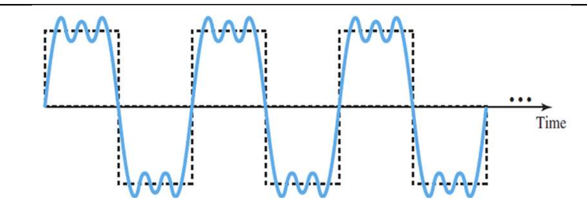


Fig: Composite signal

Sine Wave

➔ **Sine Wave**: Most fundamental periodic analog signal

- Smooth and consistent change over a cycle
- Represented by three parameters:
  - **Peak Amplitude**
  - **Frequency**
  - **Phase**

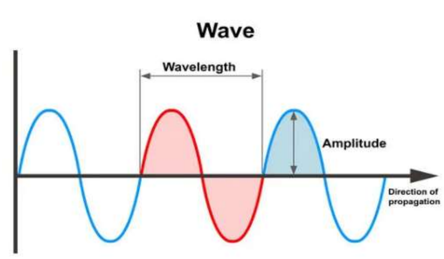
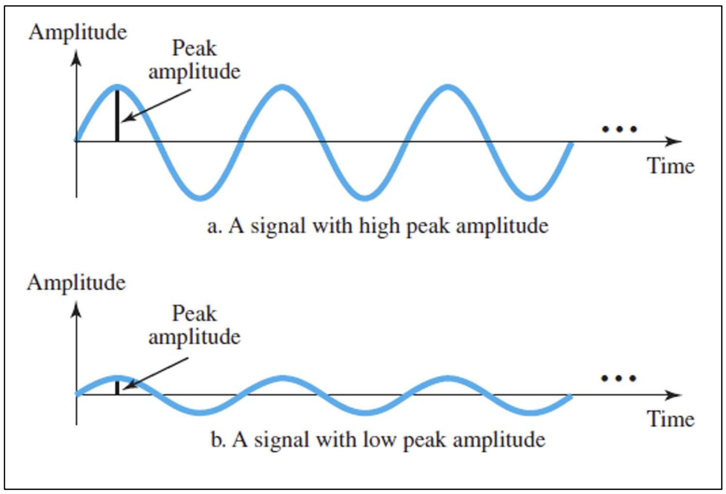
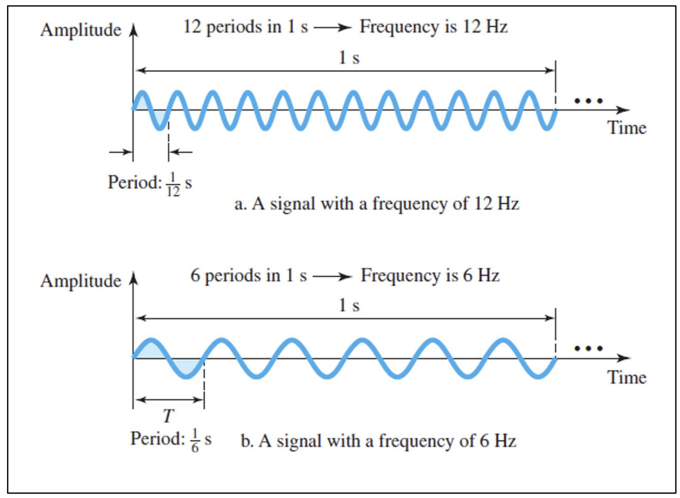


Fig: Sine wave

- **Peak Amplitude:**
  - Absolute value of highest intensity
  - Proportional to the energy carried
- **Frequency:**
  - Number of periods in 1 second (Hz = cycles per second)
  - **Period (T) & Frequency (f) Relationship**:

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$





Period and frequency

Period		*Frequency*	
Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (µs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Problem 1: The power we use at home has a frequency of 60 Hz (50 Hz in Europe). Calculate the time period in millisec?

➔ **Is given,**

f = 60 Hz

We Know,

$f = \frac{1}{T}$

$\therefore T = \frac{1}{f} = \frac{1}{60} = 16.667ms$

Problem 2: The period of a signal is 100 ms. What is its frequency in kilohertz?

➔ **Is given,**

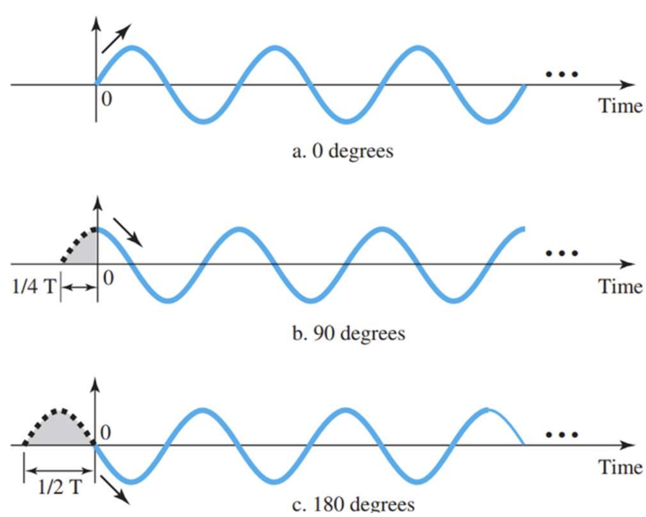
T = 100 ms

We Know,

$f = \frac{1}{T} = \frac{1}{100ms} = 10Hz = 0.01\ kHz$

Phase

- **Phase (Phase Shift)**
  - Describes the position of the waveform relative to time 0
  - Measured in **distance, time, or degrees**
- **Phase Measurement:**
  - **Degrees or Radians**
    - $\mathbf{360^\circ} = 2\pi$ rad
    - $\mathbf{1^\circ} = \frac{2\pi}{360}$ rad
    - $\boldsymbol{1\,rad} = \frac{360}{2\,\pi}$



a. 0 degrees
b. 90 degrees
c. 180 degrees

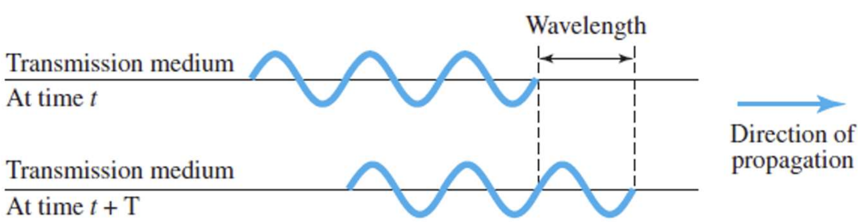
- **Phase Shifts:**
  - **360°** → Complete period shift
  - **180°** → Half-period shift
  - **90°** → Quarter-period shift

Wavelength

- **Wavelength (λ)**
  - Relates period or frequency to propagation speed
  - Distance a signal travels in one period
- **Propagation Speed**
  - Depends on **medium** and **frequency**
  - In a **vacuum**: $3 \times 10^8 \text{ m/s}$
  - **Lower** in air and even **lower** in cable

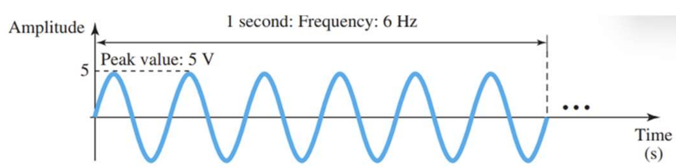
$$\textbf{Wavelength} = (\textbf{propagation speed}) \times \textbf{period} = \frac{\textbf{propagation speed}}{\textbf{frequency}}$$

$$\lambda = \frac{c}{f}$$

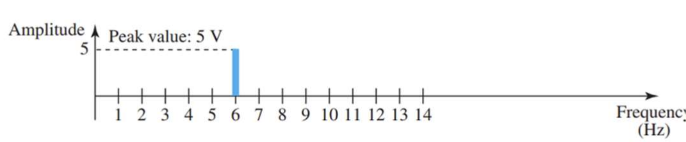


Time and Frequency Domains

- **Time-Domain Plot**
  - Displays **amplitude vs. time**
  - **Phase** is not explicitly shown
- **Frequency-Domain Plot**
  - Shows **peak amplitude vs. frequency**
  - Does **not** display amplitude changes within a period

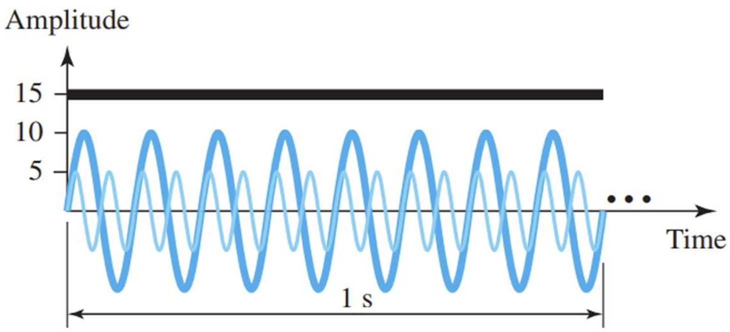


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Problem 1:

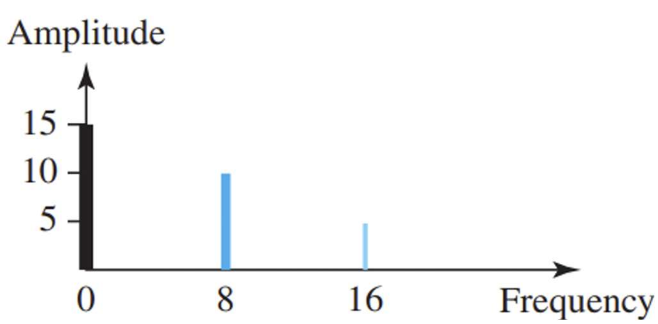


➔ Plot **Frequency Domains** where the Time-domain representation of three sine waves with frequencies 0, 8, and 16.

➔ Is given,

f1 = 0 Hz	A1 = 15
f2 = 8 Hz	A2 = 10
f3 = 16 Hz	A3 = 5

So Frequency-Domain Plot,



Composite Signal

- **Single-Frequency Sine Wave**: Not useful for data communication
- **Composite Signal**: Combination of multiple sine waves with different **frequencies, amplitudes, and phases**
- **Decomposition of Composite Signals:**
  - **Periodic Composite Signal** → Discrete frequencies
  - **Nonperiodic Composite Signal** → Continuous frequencies

❖ Periodic Composite Signal:

➔Can be decomposed into multiple signals with **discrete frequencies** in the frequency domain

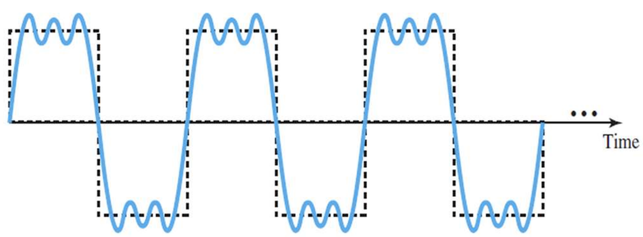


Fig: A composite periodic signal

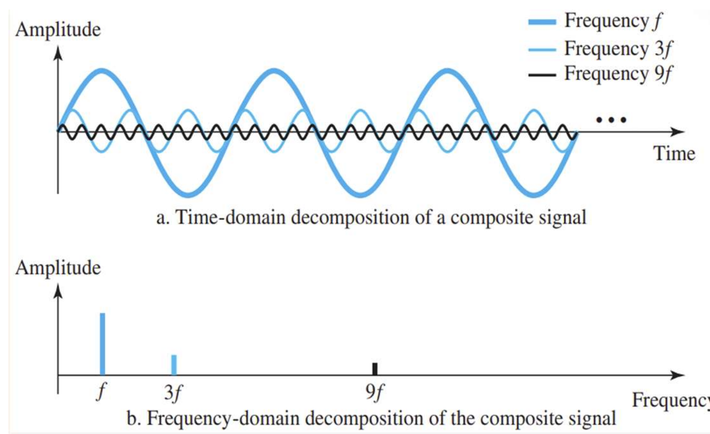


Fig: Decomposition of a composite periodic signal in the time and frequency domains

❖ Nonperiodic Composite Signal:

➔ In a time-domain representation of this composite signal, there are an infinite number of simple sine frequencies

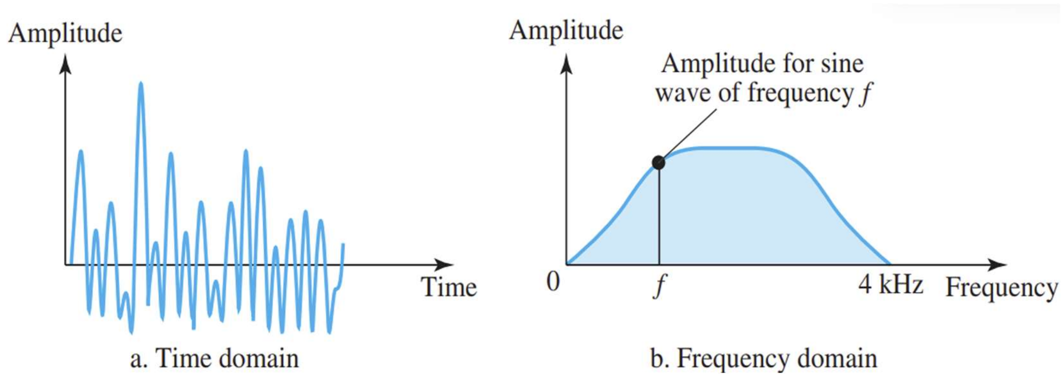


Fig: The time and frequency domains of a nonperiodic signal

Bandwidth

- The range of frequencies in a composite signal
- **Bandwidth** = Difference between the **highest** and **lowest** frequencies in the signal.

Bandwidth Formula:

$$\boldsymbol{B = f_h - f_l}$$

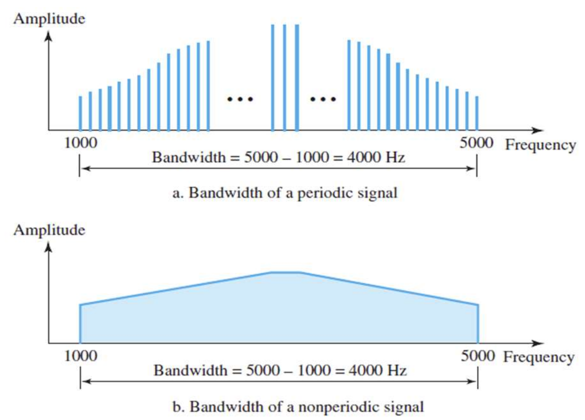


Fig: The bandwidth of periodic and nonperiodic composite signals

Problem 1: If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum(frequency), assuming all components have a maximum amplitude of 10 V.

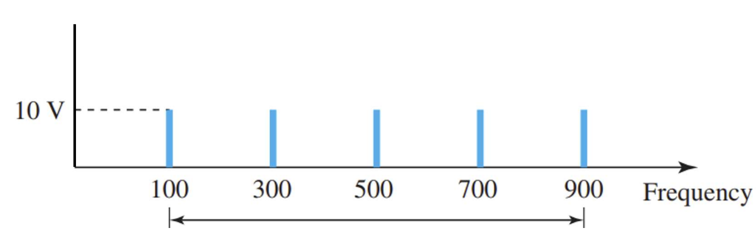
➔ **Is given,**

f1 = 100 Hz	A1 = 10
f2 = 300 Hz	A2 = 10
f3 = 500 Hz	A3 = 10
f4 = 700 Hz	A4 = 10
f5 = 900 Hz	A5 = 10

We Know,

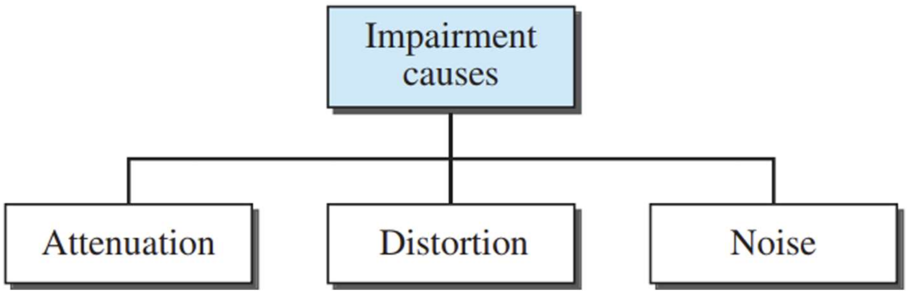
$$\boldsymbol{B = f_h - f_l}$$
$$\boldsymbol{= 900 - 100}$$
$$\boldsymbol{= 800\ Hz}$$

➔ The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz



Transmission Impairment

- **Signal Impairment**
  - Occurs because transmission media are imperfect
  - The signal at the **beginning** of the medium is not the same as the one at the **end**
  - What is **sent** is not necessarily what is **received** due to imperfections in the medium



Attenuation

- **Attenuation means:**
  - Loss of energy in a signal
  - Occurs as the signal **travels** through a medium
  - Due to **resistance** in the medium

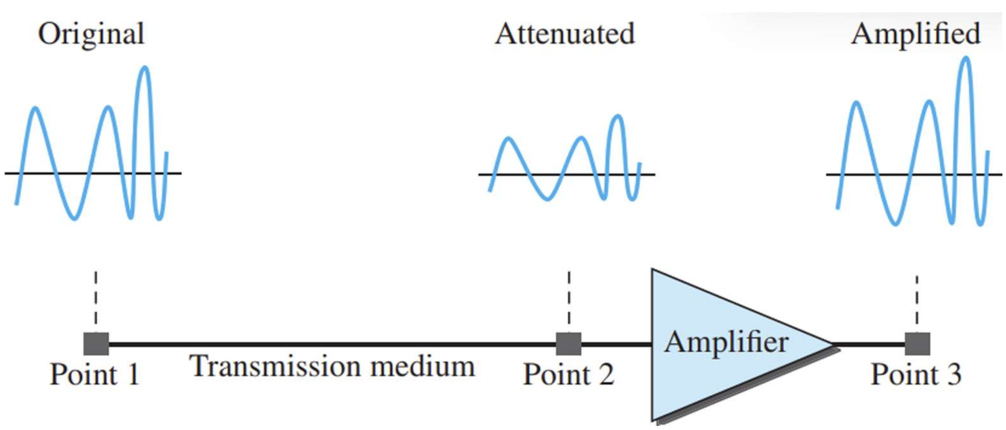


Fig: Attenuation

Decibel

- **Decibel (dB)**
  - Measures **relative strength** of two signals
  - Used to show **loss or gain** in signal strength
  - Can compare **two signals** or **one signal at different points**

$$dB = 10\ log_{10}\frac{p_2}{p_1}$$

- **Power Levels:**
  - P1 = Power at point 1
  - P2 = Power at point 2
- **Decibel (dB) Calculation:**
  - **Attenuation**: dB is **negative** (P2<P1)
  - **Amplification**: dB is **positive** (P2>P1)

Problem 1: Suppose a signal travels through a transmission medium and its power is reduced to one-half. Calculate the **Decibel value?**

➔ **Let,**

P1 = P

P2 = $\frac{1}{2}\ P$

We Know,

$$dB = 10\ log_{10}\frac{p_2}{p_1}$$

$$= 10\ log_{10}\frac{\frac{P}{2}}{P} = 10\ log_{10}\frac{P}{2}\times\frac{1}{P} = 10\ log_{10}\frac{1}{2} = -3.01\ dB$$ **(Attenuation)**

Problem 2: The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

➔ **Is given,**

dB = $-0.3\ dB/km$

P1 = $2\ mW$

∴ So,

dB at 5km = $-0.3\ \times 5\ dB = -1.5\ dB$

We Know,

$$\gg dB = 10\ log_{10} \frac{p_2}{p_1}$$

$$\gg \frac{dB}{10} = log_{10} \frac{p_2}{p_1}$$

$$\gg 10^{\frac{dB}{10}} = \frac{p_2}{p_1}$$

$$\gg P2 = 10^{\frac{dB}{10}} \times P1 = 10^{\frac{-1.5dB}{10}} \times 2\ mW = 1.4\ mW$$

Distortion

- **Distortion**
  - Occurs when a signal **changes its form or shape**
  - Affects **composite signals**
- **Cause of Distortion**
  - Each **signal component** has a different **propagation speed**
  - Leads to **varying delays** in reaching the destination
  - If delays **differ** from the period duration, **phase differences** occur

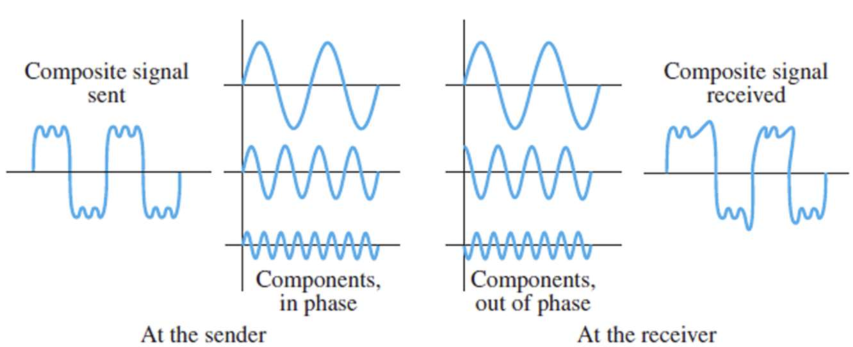


Fig: Distortion

Noise

- **Noise (Signal Impairment)**
  - Unwanted signals affecting transmission
- **Types of Noise**
  - **Thermal Noise**: Random motion of electrons in a wire
  - **Induced Noise**: From motors, appliances, etc.
  - **Impulse Noise**: From power lines, lightning, etc.

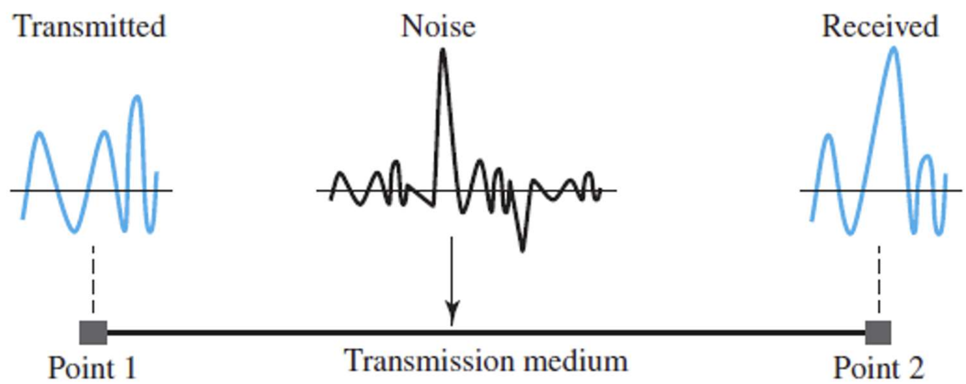


Fig: Noise

Signal-to-Noise Ratio (SNR)

- **Signal-to-Noise Ratio (SNR)**
  - Ratio of **signal power** to **noise power**
  - Represents how much of the signal is **useful** vs. **unwanted noise**
  - Defined as:

$$\boxed{\mathbf{SNR} = \left(\frac{\mathbf{Psignal}}{\mathbf{Pnoise}}\right)}$$

$$\text{Psignal} = \textit{Average Power of Signal}$$

$$\text{Pnoise} = \textit{Average Power of Noise}$$

- **SNR in Decibels (SNR_{dB})**
  - Defined as:

$$\boxed{\mathbf{SNR_{dB} = 10log_{10}SNR}}$$

  - Higher **SNR** → Better signal quality
  - Lower **SNR** → More noise, reduced clarity

Problem 1: The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}

➔ **Is Given,**

$P_{Signal} = 10$ mW

$P_{Noise} = 1$ μW

We Know,

$$\mathbf{SNR} = \left(\frac{\mathbf{Psignal}}{\mathbf{Pnoise}}\right) = \frac{10\ mW}{1\ \mu W} = 10{,}000$$

$$\mathbf{SNR_{dB} = 10log_{10}SNR} = 10 \times \log_{10} \times 10{,}000 = 40$$

Practice Problem

- **Problem 1:** Plot the **frequency domain** representation for three sine waves given in the time-domain as:

$$\mathbf{y(t1) = 20sin(40\pi t + \phi)}$$

$$\mathbf{y(t2) = 15sin(20\pi t + \phi)}$$

$$\mathbf{y(t3) = 10sin(10\pi t + \phi)}$$

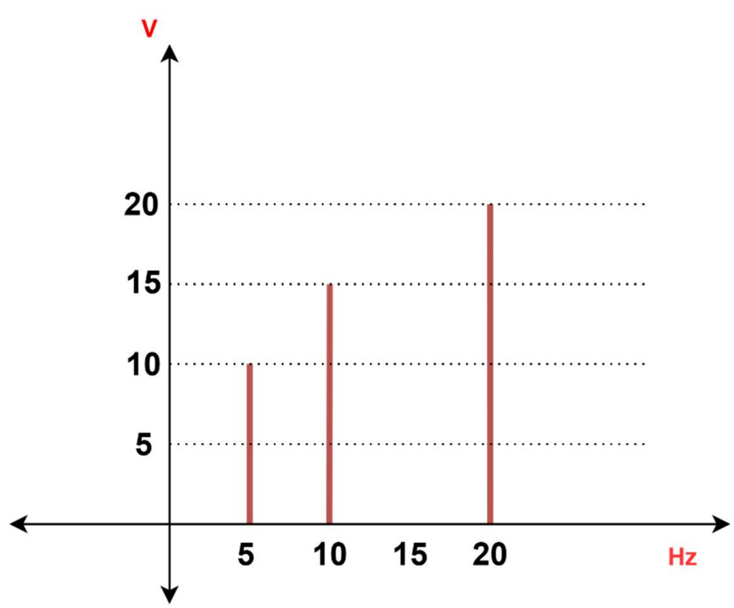
➔ **Compare the given equations with the general sine wave equation**

$$\mathbf{y(t) = Asin(2\pi ft + \phi)}$$

➔ **Is Given,**

$f1 = \frac{40}{2} = 20$ Hz	A1 = 20 V
$f2 = \frac{20}{2} = 10$ Hz	A2 = 15 V
$f3 = \frac{10}{2} = 05$ Hz	A3 = 10 V

So Frequency-Domain Plot,



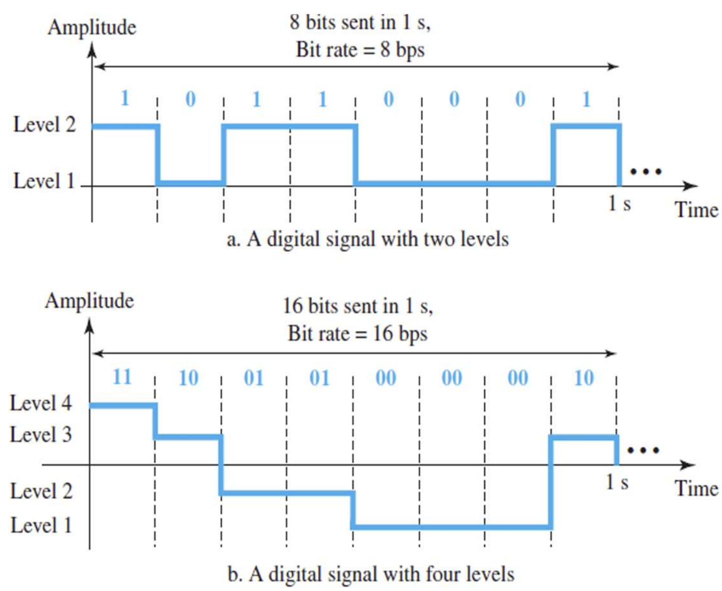
DATA RATE LIMITS

Information as Digital Signal

- **Digital Signal Representation**
  - Example: **1 → Positive Voltage, 0 → Zero Voltage**
  - Can have **more than two levels**
  - More levels allow sending **multiple bits per level**

Bits per level: $Log_2 L$

  - Example: **4 levels** → $Log_2 4$ **= 2 bits per level**



Problem 1: If each level in a digital signal carries **2 bits**, determine the total number of signal levels (**L**) required.

Is Given,

Bits per level = 2bits

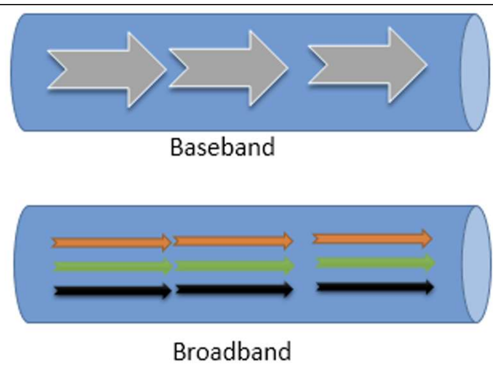
We Know,

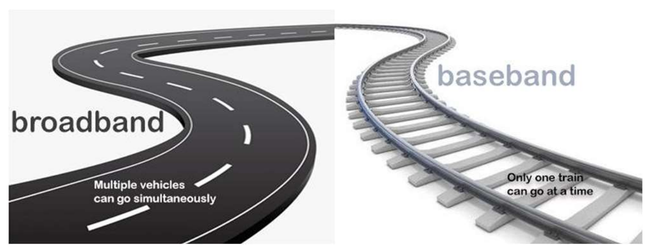
Bits per level: $Log_2 L$

$\therefore L = 2^{\text{Bits per level}} = 2^2 = 4$ **levels**

Transmission of Digital Signals

- **Digital Signal Transmission Methods:**
  - **Baseband Transmission**: Sends a **single data stream** at a time
  - **Broadband Transmission**: Sends **multiple data streams** simultaneously
- **Key Differences:**
  - **Baseband**: One channel, direct transmission
  - **Broadband**: Multiple channels, uses modulation





Differences Between Baseband and Broadband Transmission

Feature	Baseband Transmission	Broadband Transmission
Signal Type	Digital signals	Analog signals
Signal Boosting	Uses **repeaters**	Uses **amplifiers**
Data Streams	Transmits **one** stream at a time	Transmits **multiple** signals simultaneously
Communication	**Bidirectional** communication	**Unidirectional** communication
Multiplexing	Supports **TDM (Time Division Multiplexing)**	Supports **FDM (Frequency Division Multiplexing)**
Transmission Medium	Coaxial, twisted-pair, fiber-optic cables	Radio waves, coaxial, fiber-optic cables
Common Usage	Ethernet LAN networks	Cable TV, Wi-Fi, Power Line Communication

Bit Rate

- **Digital Signal Characteristics**
  - **Most digital signals are nonperiodic** → Period & frequency are **not applicable**
  - **Bit Rate**: Measures data transmission speed
  - **Definition**: Number of **bits sent per second**
  - **Unit**: **Bits per second (bps)**

- **Factors Impacting Data Rate**

The data rate (bits per second) depends on:

1. **Bandwidth Available**
2. **Signal Levels Used**
3. **Channel Quality (Noise Level)**

Formulas to Calculate Data Rate

>>Two theoretical formulas were developed to calculate the data rate:

1. Nyquist Bit Rate for a noiseless channel.
2. Shannon Capacity for a noisy channel.

Noiseless Channel: Nyquist Bit Rate

$$\boxed{\mathbf{BR = 2 \times BW \times \log_2 L}}$$

Where:

- **BR** = Bit Rate (bps)
- **BW** = Bandwidth (Hz)
- **L** = Number of signal levels

This formula provides the **theoretical maximum bit rate** for a noiseless channel.

Limitations of Increasing Signal Levels

- **Theoretical vs. Practical Limits:**
  - **In theory**, increasing the number of levels (LL) increases the bit rate.
  - **In practice**, there is a limit due to receiver complexity.
- **Receiver Burden:**
  - **For L=2L = 2 (Binary Signals):** Easy to distinguish between **0 and 1**.
  - **For L=64L = 64:** The receiver must differentiate **64 levels**, requiring higher precision.
- **Trade-off:**
  - **Higher LL → More bits per symbol** but **lower reliability**.
  - **Too many levels** introduce **errors** due to noise and hardware limitations.

Problem 1: Consider a **noiseless channel** with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What can be the maximum bit rate?

Is Given,

BW = 3000 Hz

L = 2

We Know,

$$\mathbf{BR = 2 \times BW \times \log_2 L}$$

$$\therefore \mathbf{BR = 2 \times 3000 \times \log_2 2 = 6000\ bps}$$

Problem 2: We need to send 265 kbps over a **noiseless channel** with a bandwidth of 20 kHz. How many signal levels do we need?

Is Given,

BW = 20k Hz

BR = 265k bps

We Know,

$$\mathbf{BR = 2 \times BW \times \log_2 L}$$

$$\therefore \mathbf{\frac{BR}{2 \times BW} = \log_2 L}$$

$$\therefore \mathbf{L = 2^{\frac{BR}{BW \times 2}} = 2^{\frac{265k\ bps}{20k\ Hz \times 2}} = \lceil 98.70 \rceil = 99}$$

Noisy Channel: Shannon Capacity

$$\boxed{\mathbf{C = BW \times \log_2(1 + SNR)}}$$

Where:

- **C** = Maximum channel capacity (bps)
- **BW** = Bandwidth (Hz)
- **SNR** = Signal-to-noise ratio (unitless)

➢ Key Insights:

- **No channel is truly noiseless**; every channel has some noise.
- Unlike Nyquist's formula, **Shannon's formula does not depend on signal levels (L)**.
- **No matter how many levels we use, the data rate cannot exceed the channel capacity** determined by bandwidth and noise.

Problem 1: Consider an **extremely noisy channel** in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. What is the capacity of the channel?

Is Given,

SNR = 0

We Know,

$$\mathbf{C = BW \times \log_2(1 + SNR)}$$

$$\mathbf{C = BW \times \log_2(1 + 0) = \boldsymbol{BW} \times 0 = 0\boldsymbol{bps}}$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel if SNR is zero

Problem 2: We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What is the maximum bit rate That can be achieved in this channel?

Is Given,

SNR = 63
BW = 1M Hz

We Know,

$$\mathbf{C = BW \times \log_2(1 + SNR)}$$

$$\mathbf{C = 1M \times \log_2(1 + 63) = 1\boldsymbol{M} \times \log_2(64) = 6\boldsymbol{M\ bps}}$$

Problem 3: A communication channel has a **bandwidth of 3 MHz** and a **signal-to-noise ratio (SNR) of 20 dB**.

i. Calculate the maximum theoretical data rate?
ii. If a system designer decides to operate at 80% of the Shannon capacity for better error control, what will be the actual data rate?
iii. Determine the **minimum number of signal levels (L)** required if the system operates at the chosen data rate.

Is Given,

$SNR_{dB} = 20$ $\qquad$ $\therefore SNR = 10^{\frac{SNR_{dB}}{10}} = 10^{\frac{20}{10}} = 100$
BW = 3M Hz

(i)

We Know,

$$\mathbf{C = BW \times \log_2(1 + SNR)}$$

$$\mathbf{C = 3M \times \log_2(1 + 100) = 3\boldsymbol{M} \times \log_2(101) = 19.97\boldsymbol{M\ bps}}$$

(ii)

Actual data rate at 80% of Shannon capacity:

$$\mathbf{BR_{actual} = 19.97\boldsymbol{M}\ bps \times 80\% = 15.98\ Mbps}$$

(iii)

We Know,

$$\mathbf{BR = 2 \times BW \times \log_2 L}$$

$$\mathbf{\therefore \frac{BR}{2 \times BW} = \log_2 L \gg \therefore L = 2^{\frac{BR}{BW \times 2}} = 2^{\frac{15.98M\ bps}{3M\ Hz \times 2}} = \lceil 6.33 \rceil = 7}$$

Network Performance

➢ Quality of Service (QoS) Parameters:

- **Bandwidth** – Maximum data transfer capacity.
- **Throughput** – Actual data transfer rate.
- **Latency (Delay)** – Time taken for data to travel from source to destination.
- **Bandwidth-Delay Product** – Determines the amount of data in transit at a given time.

Bandwidth

➢ Types of Bandwidth:

- **Bandwidth in Hertz** – Range of frequencies a signal or channel can carry (e.g., 4 kHz for a telephone line).
- **Bandwidth in Bits per Second** – Maximum data transfer rate of a channel or network (e.g., 100 Mbps for Fast Ethernet).

➢ Examples of Bandwidth Usage:

- **Example 1:** A **4 kHz** telephone line can transmit up to **56, Mbps** using a modem.
- **Example 2:** Increasing the bandwidth to **8 kHz** allows data transmission up to **112, Mbps** with the same technology.

Throughput

- **Throughput** measures the actual data transmission rate in a network.
- **Bandwidth** is the maximum potential data rate of a link.
- **Throughput** is usually less than or equal to bandwidth ($T \leq B$).
- **Bandwidth** is a theoretical limit, while **throughput** is the actual observed rate.

- **Example 1:**
  - **Bandwidth:** 1000 cars/minute (maximum capacity).
  - **Throughput:** 100 cars/minute (actual data transmission rate, reduced due to congestion).
- **Example 2:**
  - **Bandwidth:** 1 Mbps (maximum capacity of the link).
  - **Throughput:** 200 kbps (limited by the devices connected to the link).

Problem 1: A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Is Given,

Each frame carries = 10,000 bits
frames per minute = 12,000

∴ **Total bits per minute** = 12,000 frames × 10,000 bits/frame
= 120,000,000 bits per minute.

Now, Convert the total bits per minute to bits per second (bps):

We Know,

$$\textbf{Total bits per second (throughput)} = \frac{\textbf{Total bits per minute}}{60s}$$

$$= \frac{120{,}000{,}000}{60s} = 2\text{Mbps}$$

Latency or Delay

- **Latency (delay):** Time for a message to fully reach the destination.
- **Components of latency:**
  1. **Propagation time**
  2. **Transmission time**
  3. **Queuing time**
  4. **Processing delay**
- **Formula:**

Latency = Propagation time + Transmission time + Queuing time + Processing delay

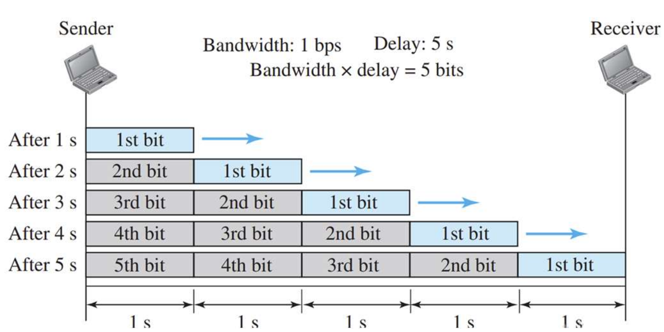
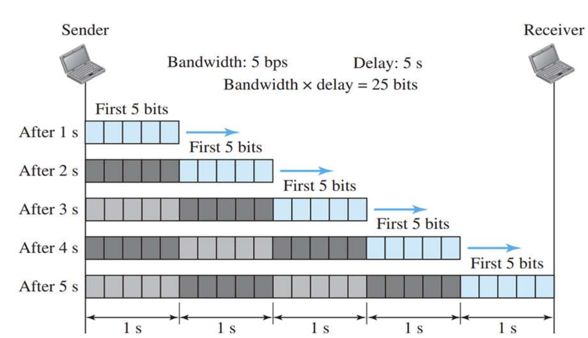
Bandwidth-Delay Product

- **Bandwidth and delay** are key performance metrics in networking.
- **Bandwidth-Delay Product (BDP):** Measures the amount of data in transit in a network.
- **Formula:**

BDP = Bandwidth × Delay

- **Represents** the capacity of a link to hold data before it reaches the destination.

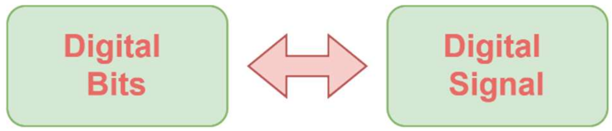
Case Study

Case1:	**Case2:**
	

- **Bandwidth-Delay Product (BDP)** = Bandwidth × Delay.
- **Case 1:** BDP = 1 × 5 = 5 bits (maximum bits in transit).
- **Case 2:** BDP = 5 × 5 = 25 bits (maximum bits in transit).
- **BDP represents** the number of bits that can occupy the link at any moment.

DATA & SIGNALS
PART-1

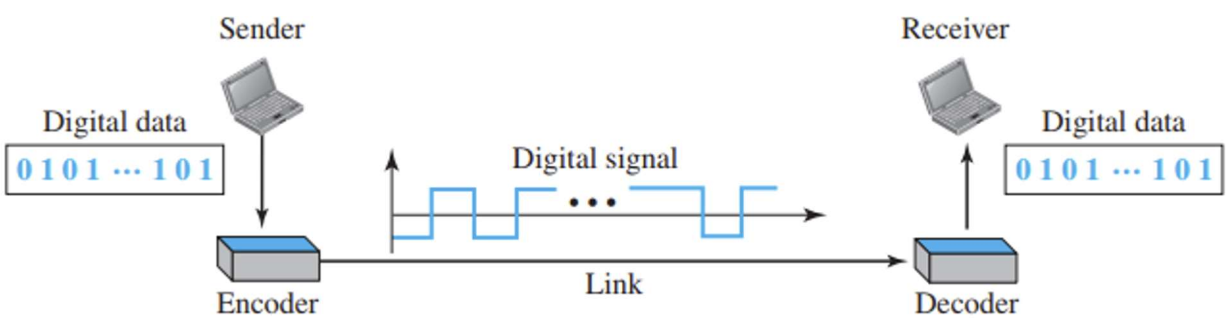
Digital to Digital Conversion



- **Digital-to-digital conversion** represents digital data using digital signals.
- Involves three techniques:
    1. **Line coding**
    2. **Block coding**
    3. **Scrambling**

Line Coding and Decoding

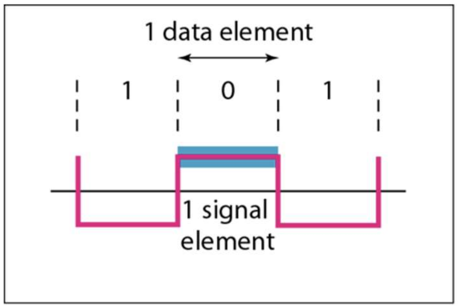
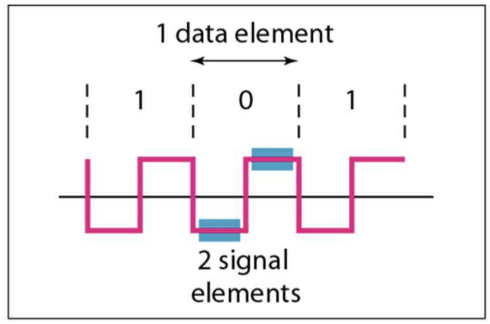
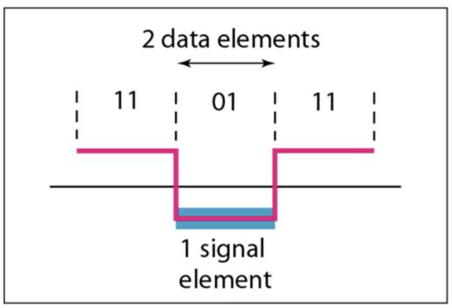
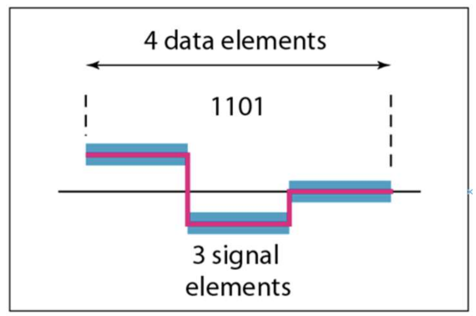
- **Line coding** converts digital data into digital signals.
- It transforms a **bit sequence** into a digital signal.
- **Sender:** Encodes data into a digital signal.
- **Receiver:** Decodes the signal back into digital data.



Signal Element vs Data Element

- **Data element:** Smallest info unit (a bit).
- **Signal element:** Smallest time unit of a digital signal.
- **Data vs Signal:**
    - Data elements = what we send
    - Signal elements = how we send
- **Ratio (r):** Number of data elements per signal element.

$$r = \frac{Data}{Signal}$$

1 	2 
$r_1 = \frac{Data}{Signal} = \frac{1}{1} = 1$	$r_2 = \frac{Data}{Signal} = \frac{1}{2} = 0.5$
3 	4 
$r_3 = \frac{Data}{Signal} = \frac{2}{1} = 2$	$r_4 = \frac{Data}{Signal} = \frac{4}{3} = 1.33$

Data Rate vs Signal Rate

- **Data rate (N):** Bits sent per second (bps).
- **Signal rate (S):** Signal elements sent per second (baud).
- **Formula:**

$$\boxed{\mathbf{S} = \mathbf{c} \times \mathbf{N} \times \frac{\mathbf{1}}{\boldsymbol{r}} \ \text{baud}}$$

Where,
S: Signal rate
N: Data rate
c: Case factor
r: Data elements per signal element

Problem 1: A signal is carrying data in which **one data element** is encoded as **one signal element** (r = 1). If the **bit rate** is 100 kbps, what is the average value of the **baud rate** if c is assumed to vary between 0 and 1?

Is Given,

Data = 1	N = 100 kbps
Signal =1	c = 0~1 = 1/2

➔ We Know,

$$\mathbf{S = c \times N \times \frac{1}{r}} \text{ baud}$$

$$S = \frac{1}{2} \times 100k \times \frac{1}{\frac{1}{1}} = 50\ kbaud$$

Problem 2: The maximum data rate of a channel is Nmax = 2 × B × log2 L (defined by the Nyquist formula). Does this agree with the previous formula for Nmax?

Is Given,

- $N_{max} = 2 \times B \times log_2 L$

➔ We Know,

$$\mathbf{S = c \times N \times \frac{1}{r}}$$

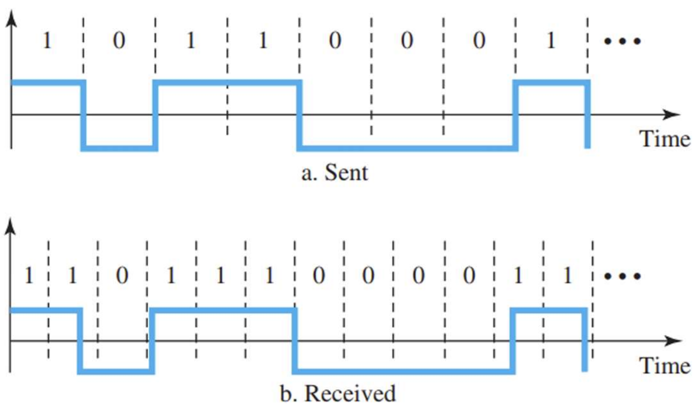
$$\gg \mathbf{N = r \times S \times \frac{1}{c}}$$

$$\therefore N_{max} = r \times S \times \frac{1}{c} = 2 \times B \times log_2 L$$

Yes, it agrees. The **Nyquist formula** shows the **maximum data rate** based on bandwidth and signal levels, while the **previous formula** relates **data rate (N)** to **signal rate (S)** and encoding efficiency. Both describe data rate limits from different perspectives.

Synchronization

- **Bit interval matching** is crucial for correct signal interpretation.
- **Clock mismatch** (faster/slower) causes misinterpretation.
- **Self-synchronization:**
  - Signal includes timing info.
  - Transitions help identify pulse timing (start, middle, end).
  - Transitions allow the receiver to **reset its clock** if out of sync.



Problem 1: In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 kbps? How many if the data rate is 1 Mbps?

Is Given,

Clock drift = 0.1% faster

Data rates = **1 kbps** and **1 Mbps**

Formula:

$$\boxed{\textbf{Extra bits per second} = \textbf{Data rate} \times \frac{\textbf{Drift}}{\textbf{100}}}$$

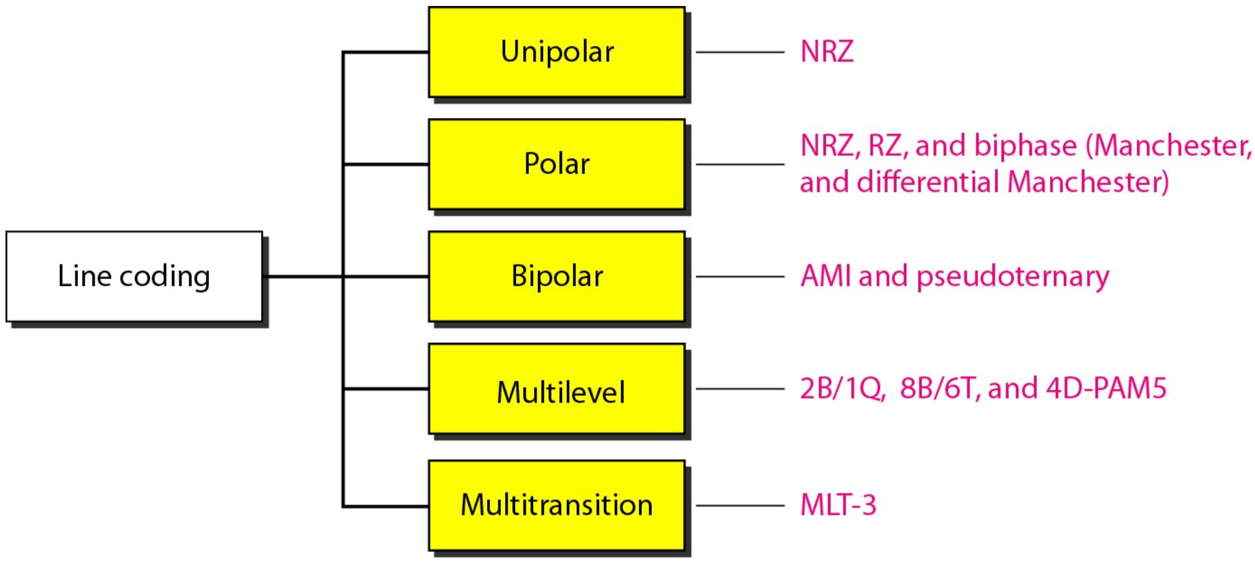
At 1 kbps:

$$\textbf{Extra bits at 1 kbps} = \mathbf{1k} \times \frac{\mathbf{0.1}}{\mathbf{100}} = \mathbf{1}\ \text{extra bit/sec}$$

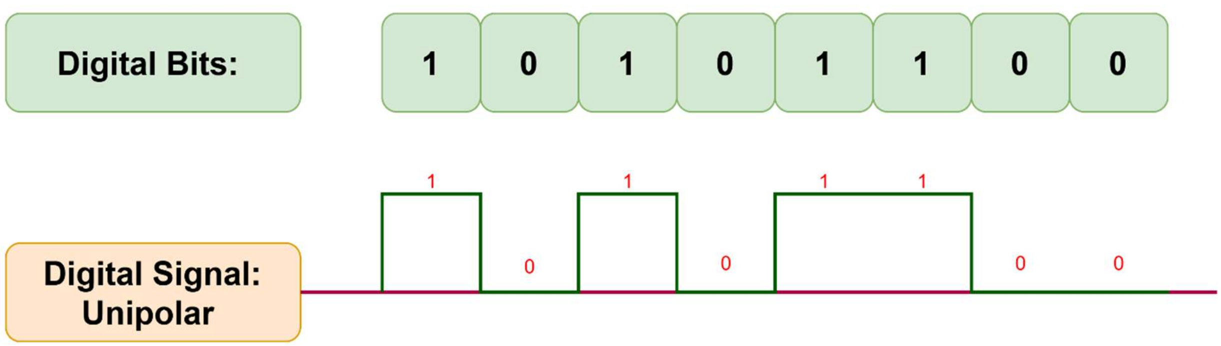
At 1 Mbps:

$$\textbf{Extra bits at 1 kbps} = \mathbf{1M} \times \frac{\mathbf{0.1}}{\mathbf{100}} = \mathbf{1000}\ \text{extra bit/sec}$$

Line Coding

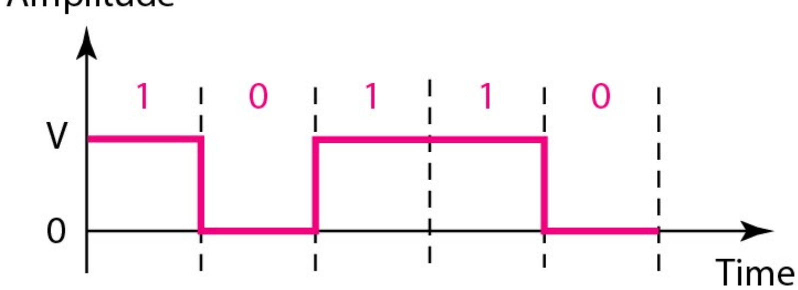


Unipolar



Non-Return to Zero (NRZ):

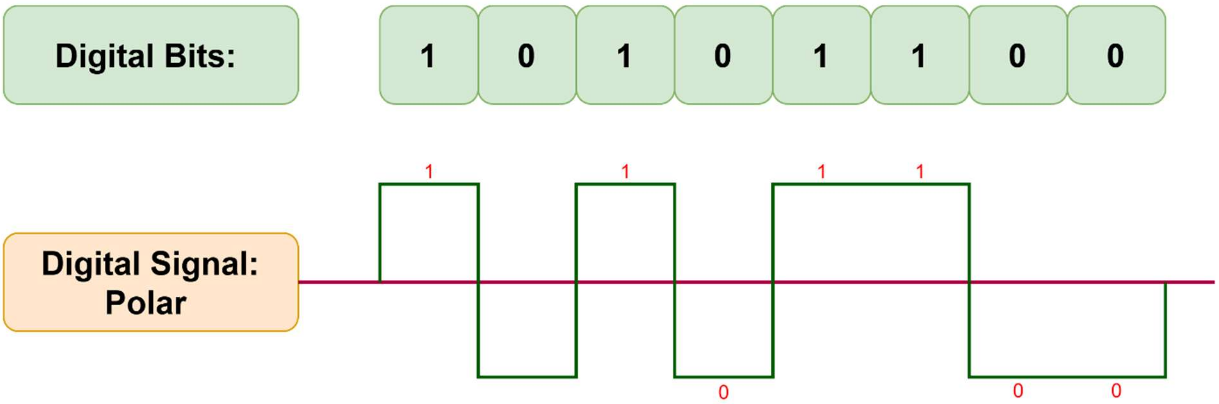
⇨ It is called NRZ because the signal does not return to zero at the middle of the bit.



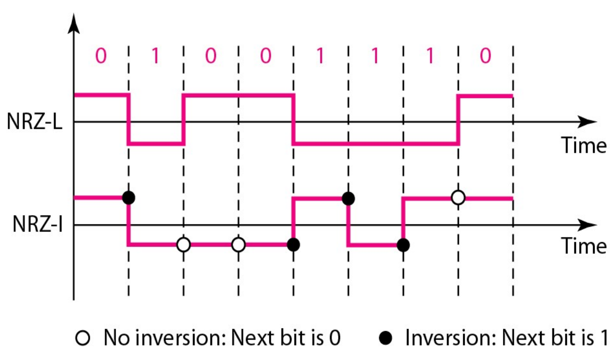
$$\frac{1}{2}V^2 + \frac{1}{2}(0)^2 = \frac{1}{2}V^2$$

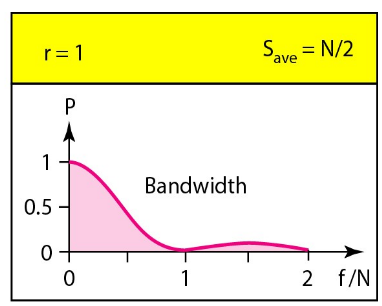
Normalized power

Polar



NRZ-L (NRZ-Level)	NRZ-I (NRZ- Invert)
0 = +Ve **1 = -Ve**	**0 = No Change** **1 = Invert**





- **NRZ-L:** Bit value depends on **voltage level**.
- **NRZ-I:** Bit value depends on **inversion or no inversion**.
- **Signal rate:** Both have an average of **N/2 Baud**.
- **Problem:** Both suffer from a **DC component issue**.

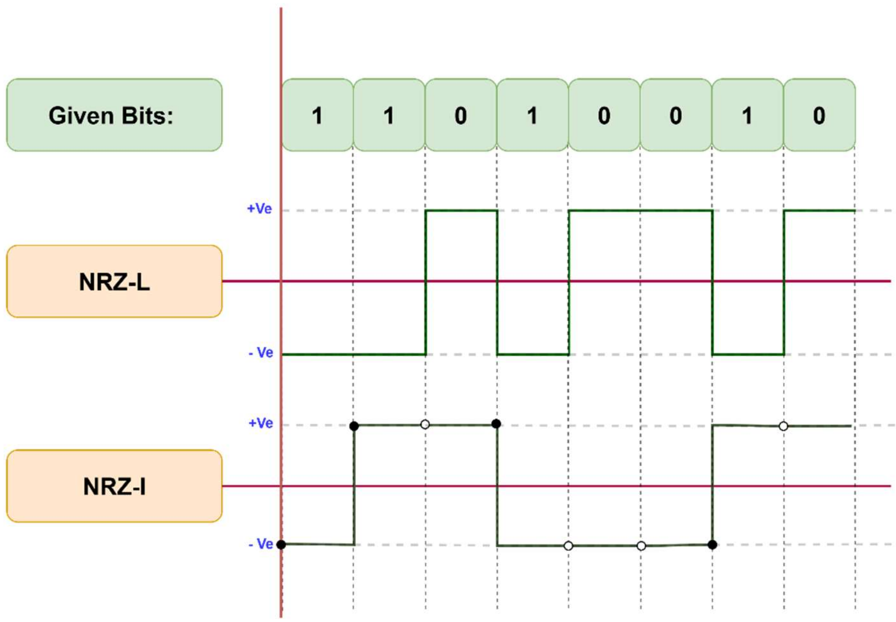
Problem 1: A system uses **NRZ-I & NRZ-L encoding** to transmit data at a rate of **2 Mbps**.

i. If the bit stream **"11010010"** is transmitted using **NRZ-I, NRZ-L** assuming the signal starts **high**, draw the signal pattern.

ii. What is the average signal rate (in Baud) for this system?

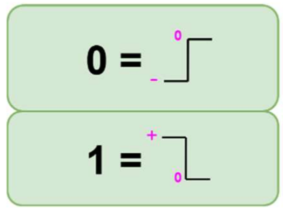
iii. If A system is using NRZ-I to transfer **10-Mbps data**. What are the average signal rate and minimum bandwidth?

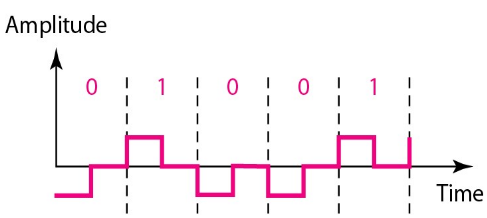
(i)

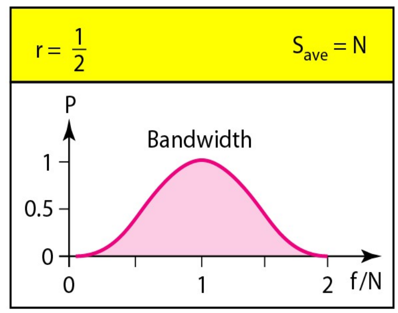


(ii)	**(iii)**
Is Given, Data = 1 N = 2 Mbps Signal =1 c = 0~1 = 1/2 ➔ We Know, $\mathbf{S = c \times N \times \frac{1}{r}}$ baud $S = \frac{1}{2} \times 2M \times \frac{1}{\frac{1}{1}} = 1 Mbaud$	**Is Given,** Data = 1 N = 10 Mbps Signal =1 c = 0~1 = 1/2 ➔ We Know, $\mathbf{S = c \times N \times \frac{1}{r}}$ baud $S = \frac{1}{2} \times 10M \times \frac{1}{\frac{1}{1}} = 5\ Mbaud$ The minimum bandwidth for this average baud rate, B_{min} = **S** = 5 MHz.

RZ: Return-to-Zero Scheme





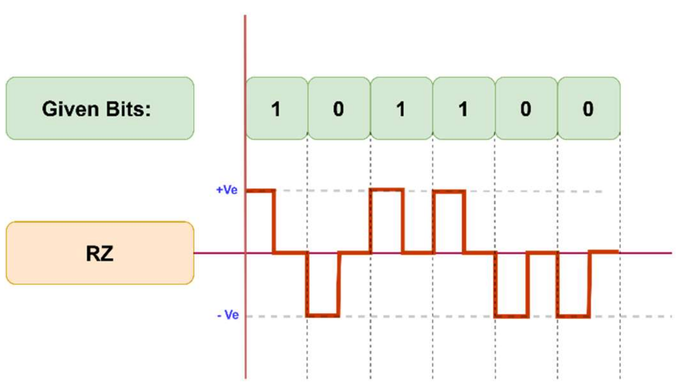


Problem 2: A data stream **"101100"** is transmitted using **RZ encoding** at a data rate of **1 Mbps**.

i. Draw the signal pattern.

ii. What is the average signal rate (in Baud) for this system?

(i)



(ii)

Is Given,

Data = 1 N = 1 Mbps

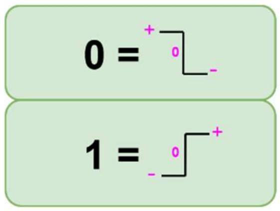
Signal =1 c = 0~1 = 1/2

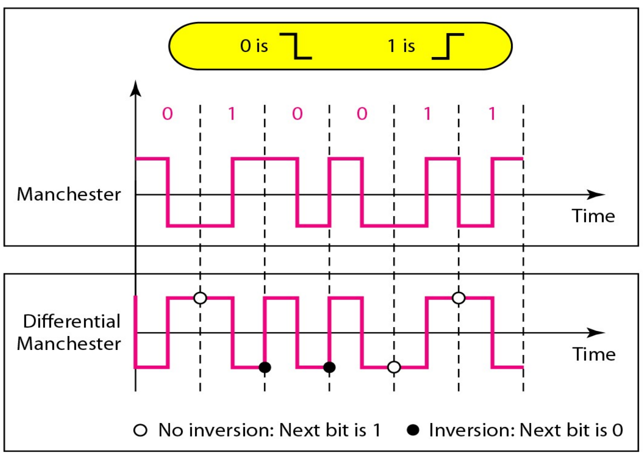
➔We Know,

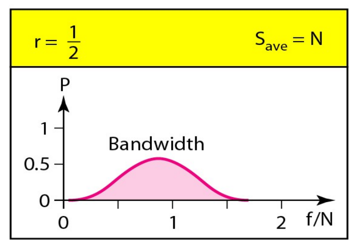
$$\mathbf{S = c \times N \times \frac{1}{r}} \text{ baud}$$

$$S = \frac{1}{2} \times 1M \times \frac{1}{\frac{1}{2}} = 1Mbaud$$

Polar Biphase

Manchester	Differential Manchester
	**0 = Invert** **1 = No Change**





- **Manchester Encoding:**
  - Transition occurs at the **middle** of the bit period for synchronization.
  - A **'1'** is represented by a **low-to-high** transition, and a **'0'** by a **high-to-low** transition.
- **Differential Manchester Encoding:**
  - Transition also occurs at the **middle** of the bit period for synchronization.
  - The **presence or absence of a transition at the beginning** of the bit determines the bit value (0 or 1).
- **Bandwidth Comparison:**
  - The **minimum bandwidth** required for both **Manchester** and **Differential Manchester** encoding is **2 times** that of **NRZ** encoding, due to the extra transitions used for synchronization.

Problem3 : A system uses **Manchester encoding** to transmit a data stream at a rate of **2 Mbps**. The data stream to be transmitted is **"101100"**.

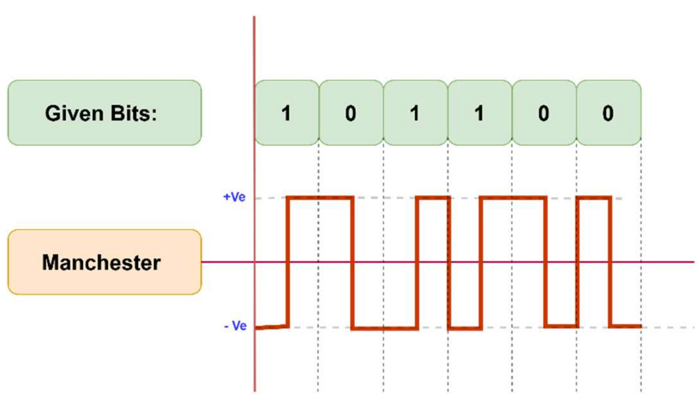
i. Draw or describe the signal waveform for the data stream **"101100"** using **Manchester encoding**.

ii. If the same data stream **"101100"** is transmitted using **Differential Manchester encoding**, describe the differences in signal representation compared to Manchester encoding.

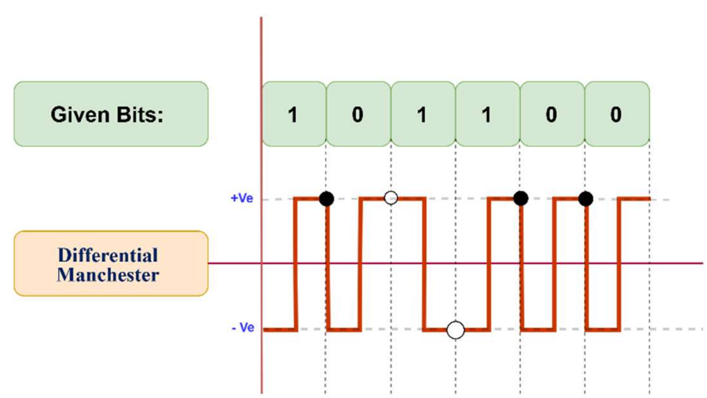
iii. What is the average signal rate (in Baud) for this system?

iv. What is the **minimum bandwidth** required for both Manchester and Differential Manchester encoding?

(i)



(ii)



(iii)

Is Given,

Data = 1　　　　N = 2 Mbps

Signal =2　　　　c = 0~1 = 1/2

→ We Know,

$$\mathbf{S = c \times N \times \frac{1}{r}} \text{ baud}$$

$$S = \frac{1}{2} \times 2M \times \frac{1}{\frac{1}{2}} = 2\ Mbaud$$

(iv)

The minimum bandwidth for this average baud rate for Manchester and Differential Manchester encoding

$$B_{\min} = \mathbf{S} = 2 \text{ MHz.}$$

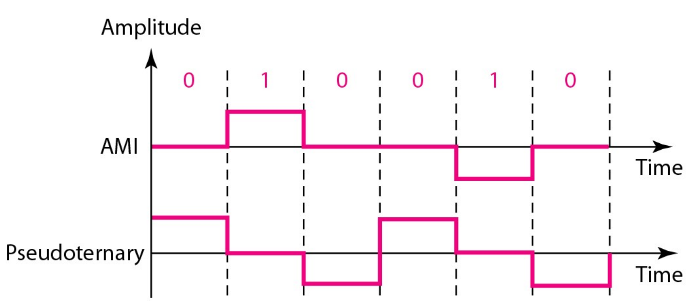
Bipolar Schemes

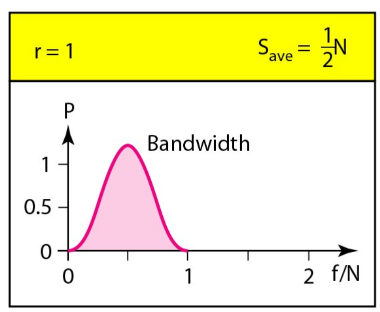
- **Bipolar Encoding (Multilevel Binary):**
  - Uses **three voltage levels**: positive, zero, negative.
  - Helps avoid **DC component** problems.

- **Two Types:**
  - **AMI (Alternate Mark Inversion):**
    - **0** → zero voltage
    - **1** → alternating **+ve** / **-ve** voltages
    - "Mark" = 1 (from telegraphy)
  - **Pseudoternary:**
    - **1** → zero voltage
    - **0** → alternating **+ve** / **-ve** voltages

- **Difference:**
  - AMI inverts **1s**,
  - Pseudoternary inverts **0s**.

AMI	Pseudoternary
0 = 0V **1 = +Ve/-Ve**	**0 = +Ve/-Ve** **1 = 0V**

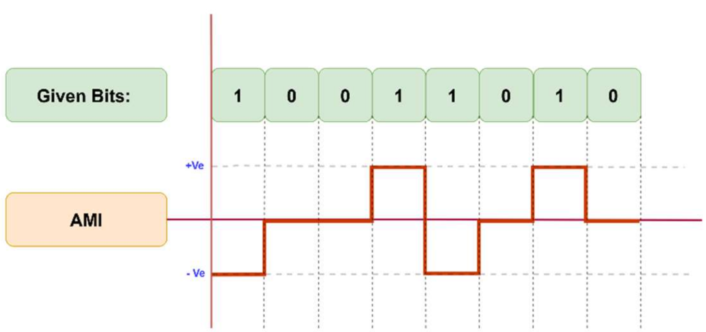




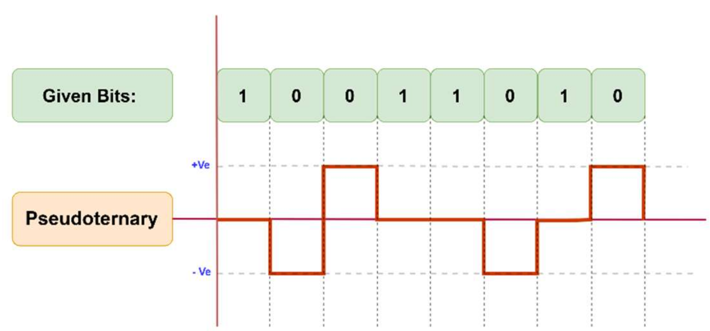
Problem4 : A system uses AMI and Pseudoternary encoding to transmit a data stream at a rate of 1 Mbps. The data stream to be transmitted **is "10011010""**

i. Draw or describe the signal waveform for the data stream **"10011010"** using **AMI encoding**, assuming the last voltage used was **+Ve**.
ii. Draw or describe the signal waveform for the same data stream using **Pseudoternary encoding**, assuming the last voltage used was **+Ve**.
iii. What is the average signal rate (in Baud) for this system?
iv. What is the **minimum bandwidth** required for both Pseudoternary and AMI encoding?

(i)



(ii)



(iii)

Is Given,

Data = 1 N = 1 Mbps

Signal =1 c = 0~1 = 1/2

➔ We Know,

$$\mathbf{S = c \times N \times \frac{1}{r}} \text{ baud}$$

$$S = \frac{1}{2} \times 1M \times \frac{1}{\frac{1}{1}} = 0.5\ Mbaud$$

(iv)

The minimum bandwidth for this average baud rate for Pseudoternary and AMI encoding

$$B_{\text{min}} = \mathbf{S} = 0.5 \text{ MHz.}$$

Multilevel Schemes

- To **increase data speed** or **reduce bandwidth**, advanced encoding schemes are used.
- Goal: Encode **m data elements** into **n signal elements**.
- **L** = number of signal levels → gives L^n **combinations**.
- The scheme is named as **mBnL coding**, where:
  - **m** = bits in binary pattern
  - **B** = Binary data
  - **n** = signal elements
  - **L** = signal levels

Examples:

- **2B1Q** → 2 bits → 1 signal element → 4 levels (quaternary)
- **4B3T** → 4 bits → 3 signal elements → 3 levels (ternary)
- **3B2B** → 3 bits → 2 signal elements → 2 levels (binary)

Redundant signal = Total combinations of Signal - Total combinations of Bits

Problem5 : A communication system uses a **4B3T** encoding scheme to transmit data.

i. What do the terms **4B3T** represent in this encoding scheme?

ii. How many **different signal patterns** are possible with 3 ternary signal elements?

iii. How many **binary combinations** can be represented by 4 bits. **(iv) find the Redundant signal**?

v. What is the **bit rate to signal rate ratio (bits per baud)** in if N= 1Mbps

(i)	**(ii)**
⇨ **4B3T**→ 4 bits → 3 signal element → 3 levels (ternary)	**Is Given,** L = 3, n= 3, **Binary** = 2, m = 4 ➔We Know, **Total combinations of Signal** = $L^n = 3^3 = 27$ **(iii)** **Total combinations of Bits** = $2^m = 2^4 = 16$
(iv)	**(v)**
➔We Know, **Redundant signal** $= L^n - 2^m = 27 - 16 = 11$	**Is Given,** Data = 4 N = 1 Mbps Signal =3 c = 0~1 = 1/2 ➔We Know, $\mathbf{S = c \times N \times \frac{1}{r}}$ baud $S = \frac{1}{2} \times 1M \times \frac{1}{\frac{4}{3}} = 0.375\ Mbaud$

Problem6 :

i. Plot or describe the digital signal waveform based on the **2B1Q** encoding for the bit pattern **'10 11 00 01'**. Assume that the previous signal level was **positive**.

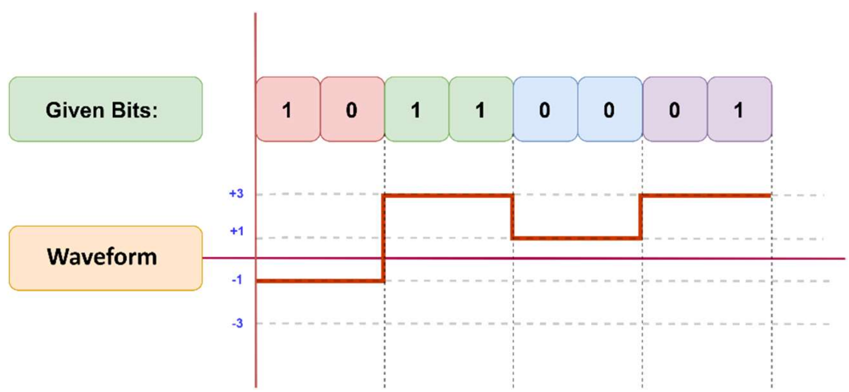
ii. What is the **bit rate to signal rate ratio (bits per baud)** in if N= 1Mbps?

iii. How many **different signal patterns** are possible?

	Previous level: positive	Previous level: negative
Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table

(i)



(ii)

Is Given,

Data = 2

Signal =1

N = 1 Mbps

c = 0~1 = 1/2

➔We Know,

$$\mathbf{S = c \times N \times \frac{1}{r}} \text{ baud}$$

$$S = \frac{1}{2} \times 1M \times \frac{1}{\frac{2}{1}} = \frac{1}{4} = 0.25\ Mbaud$$

(iii)

Is Given,

L = 4 n= 1

➔We Know,

Total combinations = $L^n = 4^1 = 4$

Problem7 :

i. Plot or describe the **digital signal waveform** for the given table below, by considering the **8B6T encoding**. Assume the signal begins at “**00010001**”.

Next bits	Weight
00010001	$-0-0++$
01010011	$-+-++0$
01010000	$+--+0+$

(i)

Weight in Multilevel Encoding:

$$W_1 = 0$$

$$W_2 = +1 = W_1 + W_2 = 0 + 1 = +1 \leq 1$$

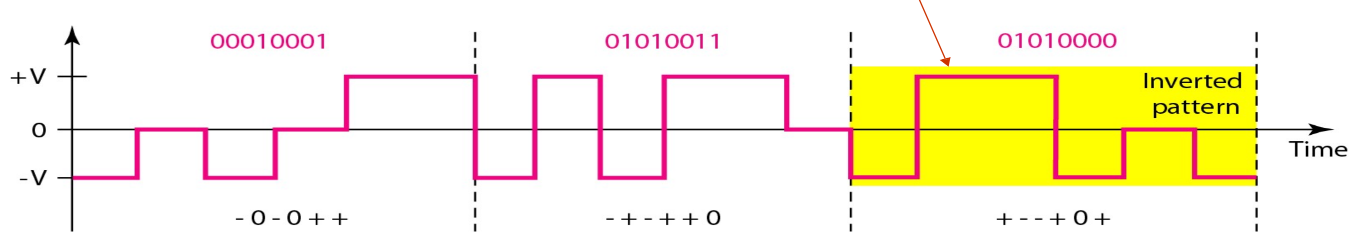
$$W_3 = +1 = W_1 + W_2 + W_3 = 0 + 1 + 1 = +2 > 1$$

$\therefore$ **Invert** $W_3 =$ **- + + - 0 - = -1**

$$SO, W_3 = -1 = W_1 + W_2 + W_3 = 0 + 1 - 1 = 0 \leq 1$$

New Weight table:

Next bits	Weight
00010001	$-0-0++$
01010011	$-+-++0$
01010000	$-++-0-$



Multiline Transmission: MLT-3

- **Signal Levels**: Uses **three levels** → +V, 0, –V
- **Transition Rules**:
    1. If next bit = **0** → **No transition**
    2. If next bit = **1** and current ≠ 0 → go to **0**
    3. If next bit = **1** and current = 0 → go to **opposite of last nonzero level**

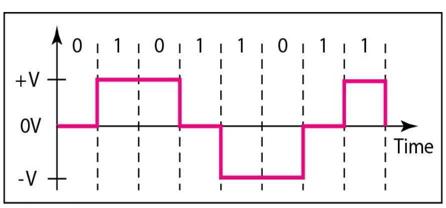
❓ Why use MLT-3?

- **Signal rate = N / 4** → only **1/4 of bit rate**
- Ideal for high-speed (e.g., **100 Mbps**) data over **copper wires**
- Reduces **electromagnetic interference (EMI)** by keeping frequency ≤ **32 MHz**

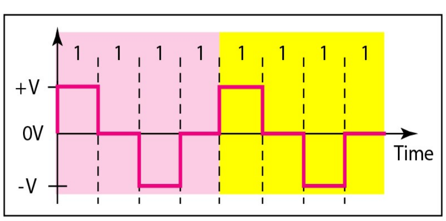
MLT-3

0 = No Change

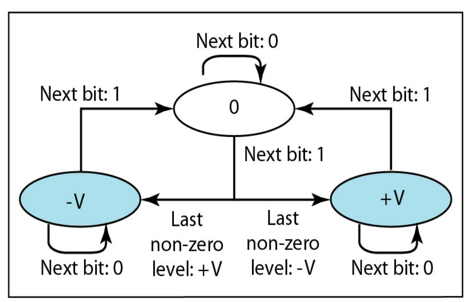
1 = +Ve/0V/-Ve



a. Typical case



b. Worse case



c. Transition states

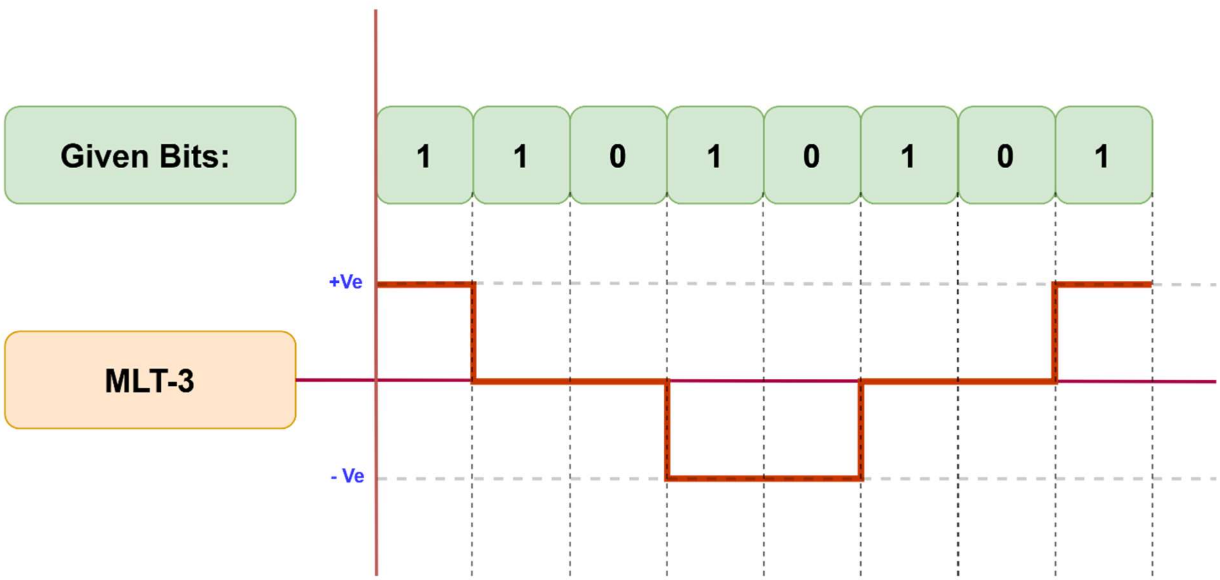
Problem 8: A system uses **MLT-3 encoding** to transmit a data stream at a rate of **100 Mbps**.

i. Given the transition rules for MLT-3, encode the data stream '11010110'. Draw or describe the signal waveform for this data stream. Assume the previous level was zero and the previous non-zero level was negative.

ii. If the data rate is **100 Mbps**, calculate the **signal rate** for MLT-3 encoding.

iii. What is the maximum frequency component present in the MLT-3 encoded signal when the bit rate is **100 Mbps**?

(i)



(ii)

Is Given,

N = 100 Mbps

→ We Know,

$$\boldsymbol{Signal\ rate} = \mathbf{N} \times \frac{\mathbf{1}}{\mathbf{4}}\ \text{baud}$$

$$Signal\ rate = \frac{100}{4} = 25\ Mbaud$$

(iii)

Since **MLT-3 encoding** results in a signal rate of **25 Mbaud** (one-fourth of the bit rate of 100 Mbps), the maximum frequency component present in the signal is **25 MHz**.

$$f_{\max} = \textbf{Signal rate} = 25\ \text{MHz.}$$

DATA & SIGNALS
PART-2

Analog to Digital Conversion

➢ Why digital signals are better than analog:

- Less affected by noise
- Can be boosted clearly using repeaters
- Easier to use with modern chips
- Can fix errors in data
- Easier to send many signals together
- Uses bandwidth more efficiently

Pulse Code Modulation (PCM)

⇨ PCM is a method used to convert an analog signal into a digital signal by sampling, quantizing, and encoding it into binary form. It is widely used in digital communication systems.

1. **Sampling** – Take snapshots of the analog signal at regular times.
2. **Quantizing** – Round each snapshot to the nearest fixed value.
3. **Encoding** – Convert the rounded values into binary numbers (bits).

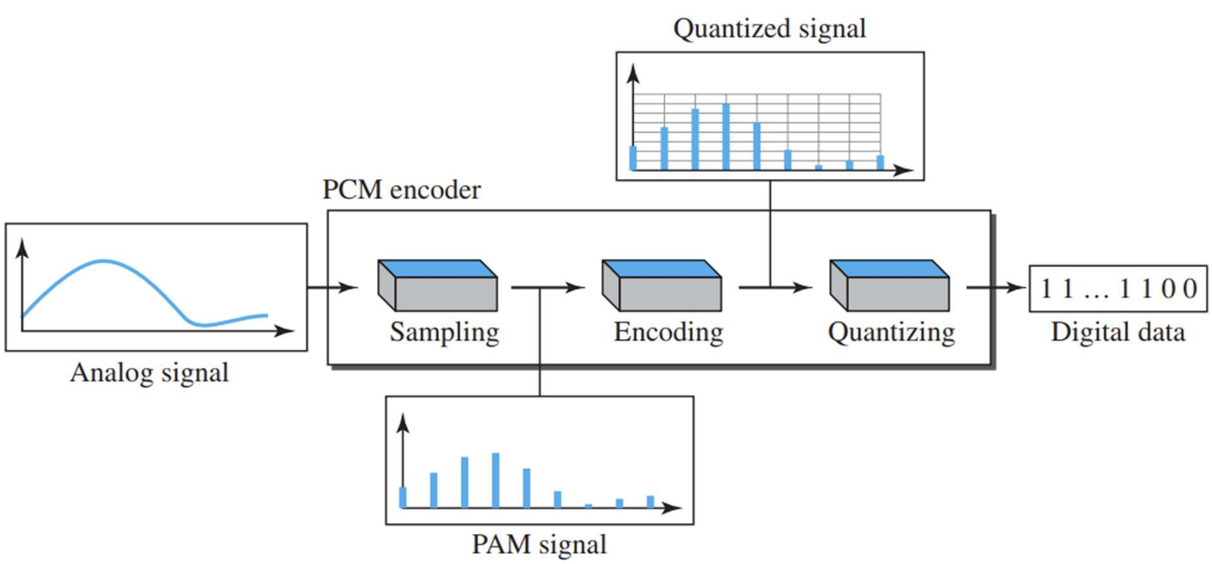


Fig: Components of PCM encoder

Sampling

- First step in PCM is sampling.
- **Sampling is done every** Ts seconds (**sampling interval**).
- **Sampling rate** $(f_s) = \frac{1}{T_s}$
- **Three sampling methods:**
  1. Ideal sampling
  2. Natural sampling
  3. Flat-top sampling

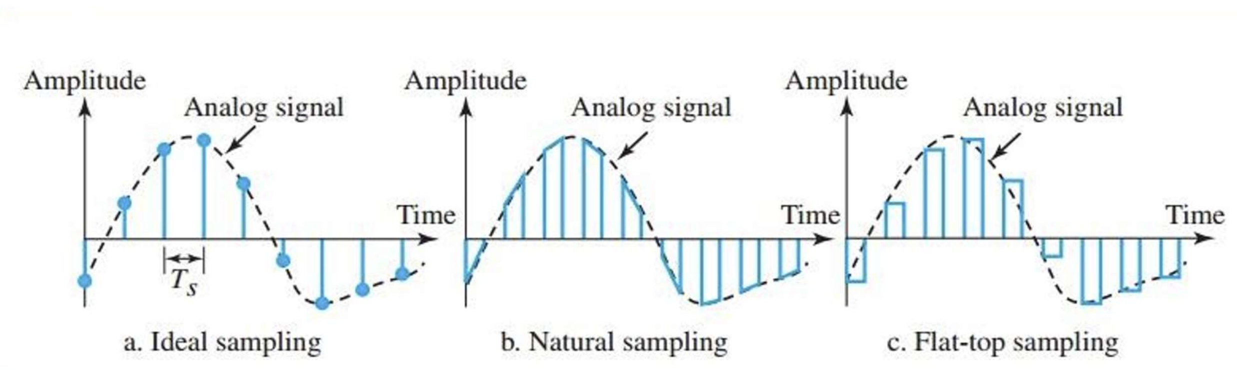


Fig: Sampling Methods

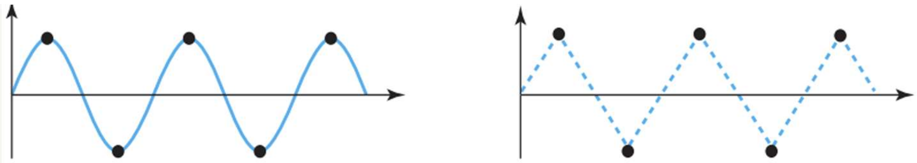
- **Ideal sampling:** Perfect but hard to implement.
- **Natural sampling:** Uses a switch; keeps the signal's shape.
- **Flat-top sampling (Sample and Hold):** Most common; creates flat pulses.
- **Sampling is also called Pulse Amplitude Modulation (PAM).**

Nyquist Theorem

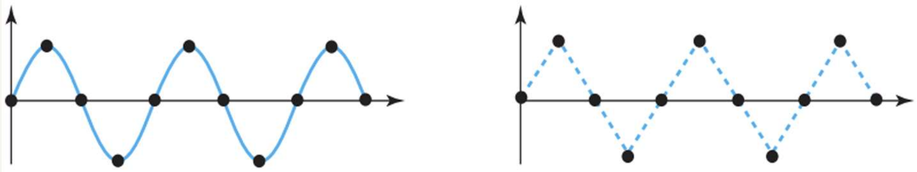
⇨ Sampling rate must be at least **2× highest frequency** in the signal.

$$\therefore \textbf{Nyquist rate} = 2 \times f_s$$

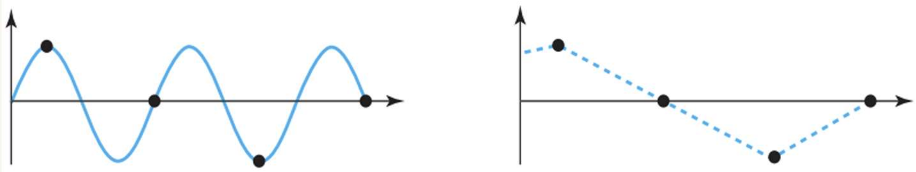
⇨ The figure below shows how a sinusoidal signal is sampled and then recovered back.



a. Nyquist rate sampling: $f_s = 2f$



b. Oversampling: $f_s = 4f$



c. Undersampling: $f_s = f$

Quantization

- **Sampling gives pulses** with amplitudes between minimum and maximum values.
- These amplitudes can be infinite and non-integer, so they **cannot be directly encoded**.
- **Quantization** converts them into a limited number of fixed levels.

➢ Steps in Quantization:

1. Signal has values between **Vmin &Vmax**.
2. Divide this range into **L levels**, each of height:

$$\boldsymbol{\Delta = \frac{(Vmax - Vmin)}{L}}$$

$$\boldsymbol{L = 2^{nb}}$$

3. Assign values **0 to L−1** to the **midpoint of each zone**.
4. Replace actual sample value with the **nearest quantized level**.

Example:

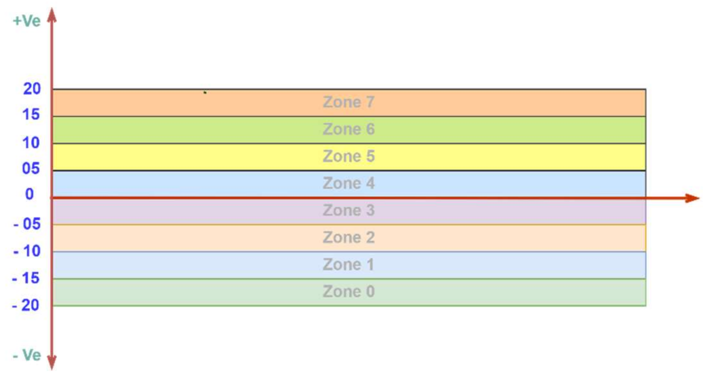
- Amplitude range: (−20V to +20V)
- **Levels (L) = 8** →

$$\Delta = \frac{(Vmax - Vmin)}{L} = \frac{\{20 - (-20)\}}{8} = 5$$

$$\boxed{\therefore \boldsymbol{No\ of\ Zone\ = L - 1}}$$

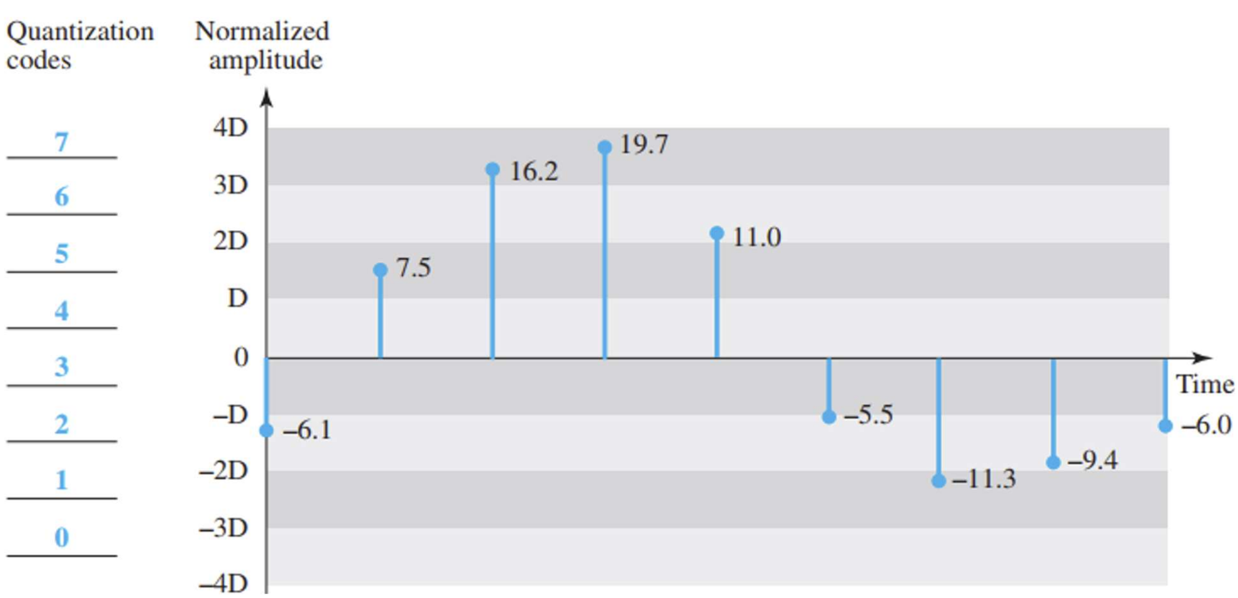
So,

$$No\ of\ Zone\ = 8 - 1 = 7$$



"**Example:** **From the figure below, find the following:**

1. Normalized PAM value
2. Normalized quantized value
3. Normalization error
4. Quantization code
5. Encoded word



➢ Normalized PAM value:

$$\boxed{\textbf{Normalized PAM} = \frac{\textbf{Sample Value}}{\boldsymbol{\Delta}}}$$

$$\text{Normalized PAM} = \frac{6.1}{5} = -1.22$$

So,

Normalized PAM values	–1.22	1.50	3.24	3.94	2.20	–1.10	–2.26	–1.88	–1.20

➢ Normalized Quantized Value:

$$\boxed{\textbf{Normalized Quantized Value} = \pm(\lfloor \textbf{Normalized PAM} \rfloor + \mathbf{0.5})}$$

$$\text{Normalized Quantized Value} = -(\lfloor 1.22 \rfloor + 0.5) = -(1 + 0.5) = -1.5$$

So,

Normalized quantized values	–1.50	1.50	3.50	3.50	2.50	–1.50	–2.50	–1.50	–1.50

➢ Normalization error:

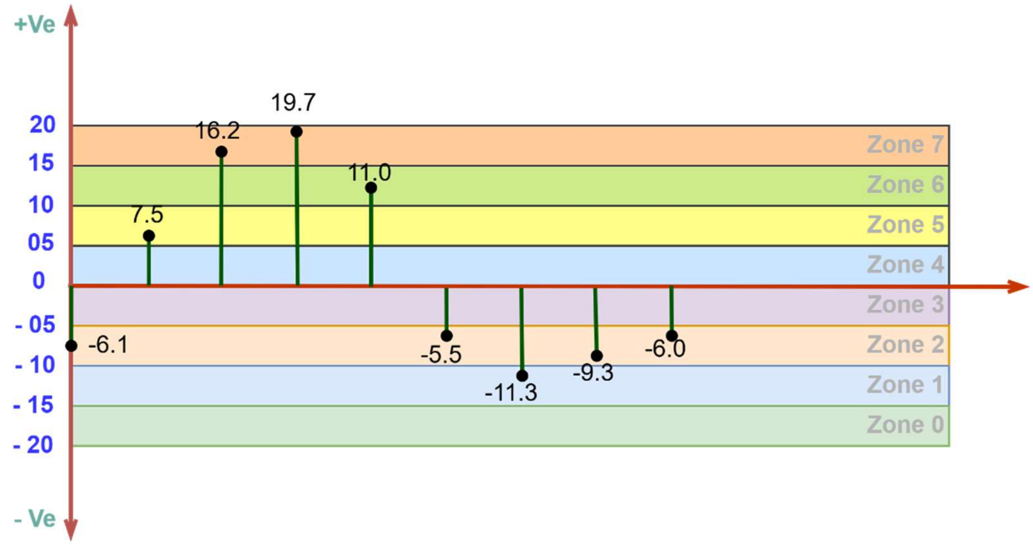
$$\boxed{\textbf{Normalization Error} = \textbf{Normalized Quantized Value} - \textbf{Normalized PAM}}$$

$$\text{Normalization Error} = (-1.5) - (-1.22) = -0.28$$

So,

Normalized error	-0.28	0	+0.26	–0.44	+0.30	–0.40	–0.24	+0.38	–0.30

➢ Quantization code:



Quantization Code = Zone number assigned to the sample value

For, -6.1:

$$\text{Quantization Code} = \text{Zone } 2 = 2$$

So,

Quantization code	2	5	7	7	6	2	1	2	2

➢ Encoded word:

Encoded Word = Binary Representation of Zone

For, 2:

$$\text{Encoded Word} = 010$$

So,

Encoded words	010	101	111	111	110	010	001	010	010

Quantization Error

Quantization Error & SNR (in dB):

- Quantization error affects **SNR (Signal-to-Noise Ratio)**.
- **SNR depends on** the number of **quantization levels (L)** or **bits per sample (nb)**.
- **Formula:**

$$\boxed{\mathbf{SNR_{dB} = (6.02 \times nb + 1.76)\ dB}}$$

Problem 1: In a PCM system, a telephone subscriber line must have an SNR (in dB) greater than 40 dB. What is the **minimum number of bits per sample** required to achieve this?

Is Given,

$SNR_{dB} = 40$

➔We Know,

$$\mathbf{SNR_{dB} = 6.02 \times nb + 1.76}$$

$$\mathbf{nb} = \frac{SNR_{dB} - 1.76}{6.02} = \lceil 6.35 \rceil = 7$$

Encoding

- After quantization, each sample is converted into an **nb-bit code word**.
- **nb** (bits per sample) is based on number of quantization levels **L**:

$$\boxed{nb = log_2 L}$$

- The **bit rate** of the PCM system can be calculated as:

$$\boxed{\mathbf{Bit\ Rate = nb \times} \boldsymbol{Nyquist\ rate}}$$

or

$$\boxed{\mathbf{Bit\ Rate = nb \times} \boldsymbol{fs}}$$

Problem 2: We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

Is Given,

nb = 8, human voice frequency = 0 to 4000 Hz

➔We Know,

$$\textbf{Bit Rate} = \mathbf{nb} \times \boldsymbol{fs} \times \mathbf{2}$$

$$\text{bit Rate} = 8 \times 4000 \times 2 = 64 \text{ kbps}$$

Bandwidth

- **Bit rate** of a PCM signal:

$$\boldsymbol{r = c \times fs \times nb}$$

- If $\boldsymbol{fs = 2 \times B_{analog}}$ (from Nyquist), then:

$$\boxed{\boldsymbol{r = c \times 2 \times B_{analog} \times nb}}$$

- **Minimum bandwidth** of the line-coded signal:

$$\boxed{\boldsymbol{B_{min} = \frac{r}{2}}}$$

- If $\boldsymbol{c = \frac{1}{2}}$ and using NRZ or bipolar encoding ($\boldsymbol{r = \frac{1}{1}}$), then:

$$\boxed{\boldsymbol{B_{min} = nb \times B_{analog}}}$$

The digital signal requires a minimum bandwidth that is nb times larger than the analog signal's bandwidth. This is the trade-off for digitizing the signal.

Problem 3: We have a low-pass signal with a bandwidth of 200 KHz that needs to be digitalized using **pulse code modulation**. Consider the sampling frequency is 700 KHz and has 1024 levels of quantization. Find

a. bit rate.
b. SNR_{dB}
c. bandwidth of the converted/digital signal.

(a)

Is Given,

sampling frequency fs = 700Hz, L = 1024

➔ We Know,

$$\textbf{Bit Rate} = \textbf{nb} \times \boldsymbol{fs}$$

$$\textbf{Bit Rate} = \boldsymbol{log_2 L} \times \boldsymbol{fs}$$

$$\text{bit Rate} = log_2 1024 \ \times 700\text{k} = 7 \text{ Mbps}$$

(b)

Is Given,

sampling frequency fs = 700 KHz, L = 1024, B_{analog} = 200 KHz

➔ We Know,

$$\textbf{SNR}_{\textbf{dB}} = \textbf{6.02} \times \textbf{nb} + \textbf{1.76}$$

$$\textbf{SNR}_{\textbf{dB}} = \textbf{6.02} \times \boldsymbol{log_2}\textbf{L} + \textbf{1.76}$$

$$\text{SNR}_{\text{dB}} = 6.02 \times log_2 1024 + 1.76 \text{ db} = 61.96 \text{ db}$$

(c)

➔ We Know,

$$\boldsymbol{B_{min}} = \boldsymbol{nb} \times \boldsymbol{B_{analog}}$$

$$B_{min} = 10 \times 200k = 2\ MHz$$

Problem 3: We have a low-pass signal with a bandwidth of 200 KHz that needs to be digitalized using **pulse code modulation**. Consider has 1024 levels of quantization. Find

a. sampling frequency
b. bit rate.
c. SNR_{dB}
d. bandwidth of the converted/digital signal.

(a)

Is Given,

$B_{analog} = 200$ KHz,

➔We Know,

$$fs = 2 \times B_{analog} = 2 \times 200k = 400kHz$$

(b)

Is Given,

sampling frequency fs = 700Hz, L = 1024

➔We Know,

$$\textbf{Bit Rate} = \textbf{nb} \times \boldsymbol{fs}$$

$$\textbf{Bit Rate} = \boldsymbol{log_2 L} \times \boldsymbol{fs}$$

$$\text{bit Rate} = log_2 1024 \times 400\text{k} = 4 \text{ Mbps}$$

(c)

Is Given,

sampling frequency fs = 400 KHz, L = 1024, $B_{analog} = 200$ KHz

➔We Know,

$$\mathbf{SNR_{dB} = 6.02 \times nb + 1.76}$$

$$\mathbf{SNR_{dB}} = \mathbf{6.02} \times \boldsymbol{log_2}\mathbf{L} + \mathbf{1.76}$$

$$\text{SNR}_{\text{dB}} = 6.02 \times log_2 1024 + 1.76 \text{ db} = 61.96 \text{ db}$$

(d)

➔We Know,

$$\boldsymbol{B_{min} = nb \times B_{analog}}$$

$$B_{min} = 10 \times 200k = 2\ MHz$$